



Kunak Air Pro Calibration

Application Note

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1. GAS AND PARTICLE CALIBRATION

Below, it is explained how to make a correct gas and particle calibration of the **Kunak Air** device. The calibration consists of adjusting the sensor's parameters based on the relation between the concentration value entered by the user and the concentration measured by the sensor itself. Values entered by the user should come from valid sources like official air quality stations or reference certified instrument.

To facilitate the **Kunak Air** device calibration, based on the reference value entered by the user, two tools are provided in "Operation" tab > "Gas & Particle Calibration" in the **KunakCloud** section.

1.1 Gas calibration

Kunak Air device integrates electrochemical sensors to measure gas pollutants. It is possible to calibrate the gas sensors by the "Baseline and Span calibration" in the "Gas Calibration" tool that can be found in "Operation" tab > "Gas & Particle Calibration" in the **KunakCloud** section.

- **Baseline:** it adjusts the sensor's zero conditions, the sensor measurements for null or minimum gas concentrations.
- **Span:** it increases or reduces the measurement range by adjusting the sensor's sensitivity.

In order to ensure the reliability of the **Kunak Air** sensor readings, the performance of the sensors should be periodically checked or calibrated. The calibration periodicity depends on the sensor properties, environmental conditions and the user/application requirement for the accuracy and reliability of the readings.

To perform the sensor calibration, there are a few options for the user.

1. Outdoor calibration
 - a. With reference station: co-location of the devices at an official reference station.
 - b. With GOLD device: In the "golden sample" approach a compact sensor recently and properly evaluated at a reference station is circulated at sensor sites for comparison of readings.
 - c. Without reference: only the baseline of the sensors can be calibrated.
2. Laboratory calibration



NOTE: A factory calibration is always carried out by Kunak to assure the quality and performance of every Kunak Air device. At least, every unit has been working continuously for 5 days, calibrated and compared to a reference unit before shipping

1.1.1 Outdoor calibration

Please, note that every **Kunak Air** has been laboratory calibrated and the calibration has been validated before shipping at the outdoor environmental conditions before the shipping. Any big change in the environmental conditions (+/-10°C average temperature) may slightly affect factory calibration settings.

1.1.1.1 With reference station

Place the **Kunak Air** device near an official reference station to carry out a colocation test. For the most accurate calibration, place the **Kunak Air** at a maximum distance of 2 meters from the inlet port of the reference air quality station.

Once **Kunak Air** has been installed, wait for at least 24h for stabilization before comparing any data between **Kunak Air** and the reference station. Then, if required, calibrate **Kunak Air** as shown in sections 1.1.3 and 1.1.4.

After calibration, keep on the device working for several hours or days to compare the data with the reference station to guarantee a correct performance, as specified in the datasheet and/or shown in the field tests that can be found in the "Support" section of the **KunakCloud** web.

Once the calibration has been validated, switch off the unit. Now, **Kunak Air** is ready to be deployed anywhere. If required, Kunak can integrate¹ into **KunakCloud** platform any third-party data (for example, air quality reference stations data) to make easier the calibration process.

1.1.1.2 With GOLD device

Place the GOLD device near the **Kunak Air** device to be calibrated. For the most accurate calibration, place the **Kunak Air** and the GOLD device at a maximum distance of 1 meter between them.

Once they have been installed, wait for at least 24h for stabilization before comparing any data between **Kunak Air** and the GOLD device. Then, if required, calibrate the **Kunak Air** as shown in sections 1.1.3 and 1.1.4.

After calibration, keep on the device working for several hours or days to compare the data with the GOLD device to guarantee a correct performance as specified in the datasheet and/or shown in the field tests that can be found at the "Support" section of the **KunakCloud** web.

Once the calibration has been validated, switch off the GOLD unit. Now, the GOLD device can be moved to another location to calibrate another device.

1.1.1.3 Without reference

If there is no access to any reference data, a rough "calibration/adjustment" can be done in the baseline of the sensors.

In this case, please contact Kunak for further and specific assistance. Mainly, gases will usually reach a minimum concentration during a certain part of the day. For example, it is possible to find concentrations that vary, depending on the type of gas, from zero ppb to a few ppb at night time. The user must find these periods to perform a correct baseline adjustment or correct negative night values as a zero. We encourage our customers to contact Kunak if they want to carry out this type of calibration.

If calibration is carried out with the correct parameters, **Kunak Air** is expected to have the same correlation (R^2) and accuracy values as the ones specified in the datasheets.

If the user is looking for the maximum accuracy, a new baseline calibration is required if the average temperature changes in 10 degrees from the average temperature during calibration.

1.1.2 Laboratory calibration

1.1.2.1 Initial Set Up

It is advisable to use one of the settings described below to ensure the quality of the laboratory tests.

1. Sealed enclosure: Composed of a sealed enclosure with a controlled input and output. In the middle of the enclosure must be positioned the DUT (Device Under Test) **Kunak Air**. Below the sensor surface, a sample probe must be connected to reference equipment to guarantee the real concentration around the sensor area. The maximum distance from the surface of the sensor and the input of the reference equipment must be lower than 5 cm.

¹ The data of the reference station to integrate in the KunakCloud must be public and correctly posted on the internet or a place that it can be accessible to Kunak Software to collect the data.

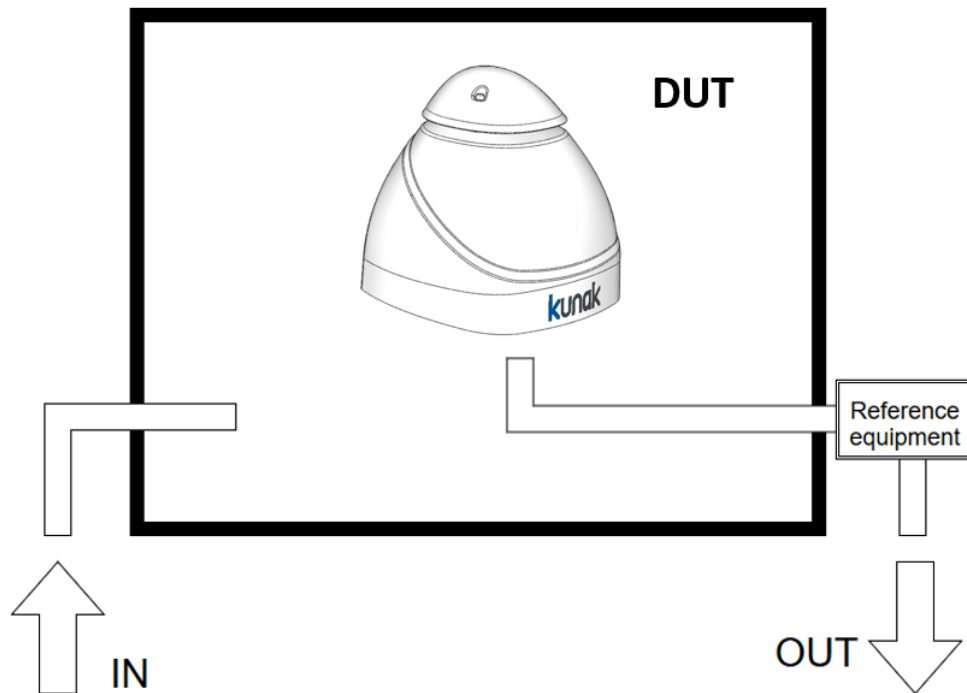


Figure 1 – Sealed enclosure

2. **Gas-Hood:** Design a gas hood that fits under **Kunak Air** with an input and output port. The gas hood is required to enclose the gas sensors and the humidity/temperature sensor. The temperature and humidity sensor is important to the **Kunak Air** device because the gas sensors are affected by these parameters and the device corrects the measurements obtained by using the temperature and humidity values. It must be taken into account any possible reaction that can be produced by the contact of the gas and the tool, so inert materials are mandatory for any gas hood designs.



NOTE: It must be considered that it can be losses of gas through the humidity and temperature sensor as well as any loss within the material's surface. Kunak strongly recommend the customer to always take the output of the tool into reference equipment to know the real concentration on the sensor surface

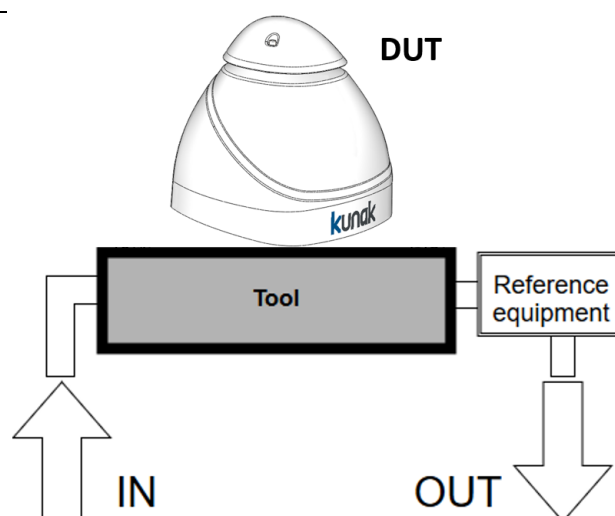


Figure 2 – Gas Hood

The temperature, humidity and pressure conditions for the tests should be set like the ones that the device will have in its final location. If these data are unknown, the recommended conditions are the ones described in Table 1 – Temperature, humidity and pressure condition for the test.

Parameter	Operating Range	Recommended Testing Values
Temperature	-30 °C to 40 °C	20 °C
Humidity	15 %Rh to 85 %Rh	50 %Rh
Pressure	80 KPa to 120 KPa	101.325 KPa

Table 1 – Temperature, humidity and pressure condition for the test

CAUTION!

Temperature and humidity affect the sensors. High changes in temperature or humidity affect the gas sensors readings:



- Prolonged exposure to humidity above 85% RH can damage the gas sensors.
- The temperature and humidity must be as stable as possible during the calibration procedure.
- Changes may not exceed ± 5 °C per hour and $\pm 10\%$ RH per hour for a correct laboratory test or calibration.
- If there is an abrupt change in temperature or humidity, the sensor may require 4 hours for stabilization.

1.1.2.2 Calibration procedure

Initial considerations before starting the calibration procedure:

- Configure the sampling period of the sensors to test **Kunak Air** to measure every 30 seconds or 1 minute and the sampling period to 30 minutes or more to have enough data points. To modify the sampling period remotely for each sensor, go (in the **KunakCloud** website) to “Configuration > Simple > Sensor” and select the device and the sensor to modify. Specify the sampling period for that sensor at “Sampling > Period” and click on “Save” button to send it to the device. In the next sending period of the device, it will be changed.

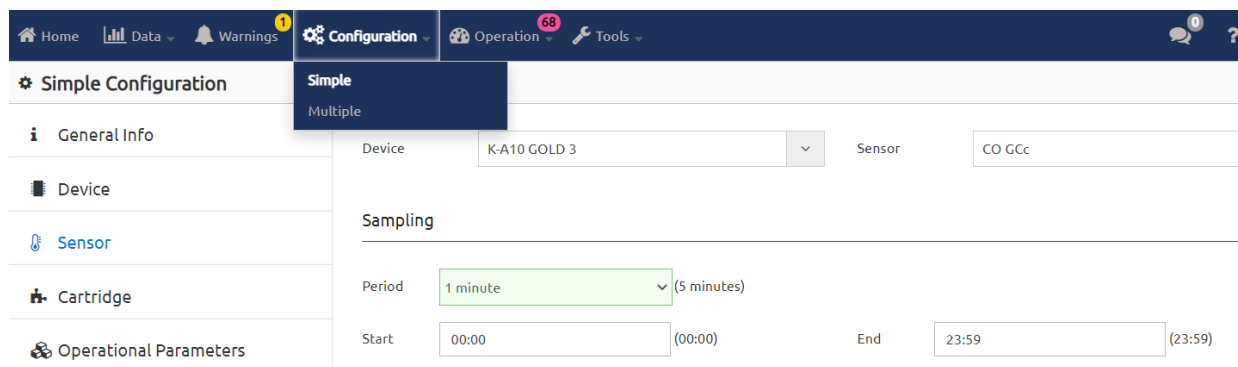


Figure 3 – Change of the device sampling period

- Configure the sending period taking into account that the equipment will stop measuring from 1 minute to 5 minutes maximum (depending on sampling period and sending period), after sending the values to the **KunakCloud**, to avoid electromagnetic interference in the gas sensors. Electromagnetic interferences of the cellular communications can affect the sensor readings. It is not recommended to configure the sending period of more than 10 min to carry out these tests. To change the sending period remotely go, on the **KunakCloud** website, to “Configuration > Simple > Device”. Select the device to modify the sending period and modify it in “Communications > Sending period” and click on “Save” button to send it to the device. In the next sending period of the device, will be changed.

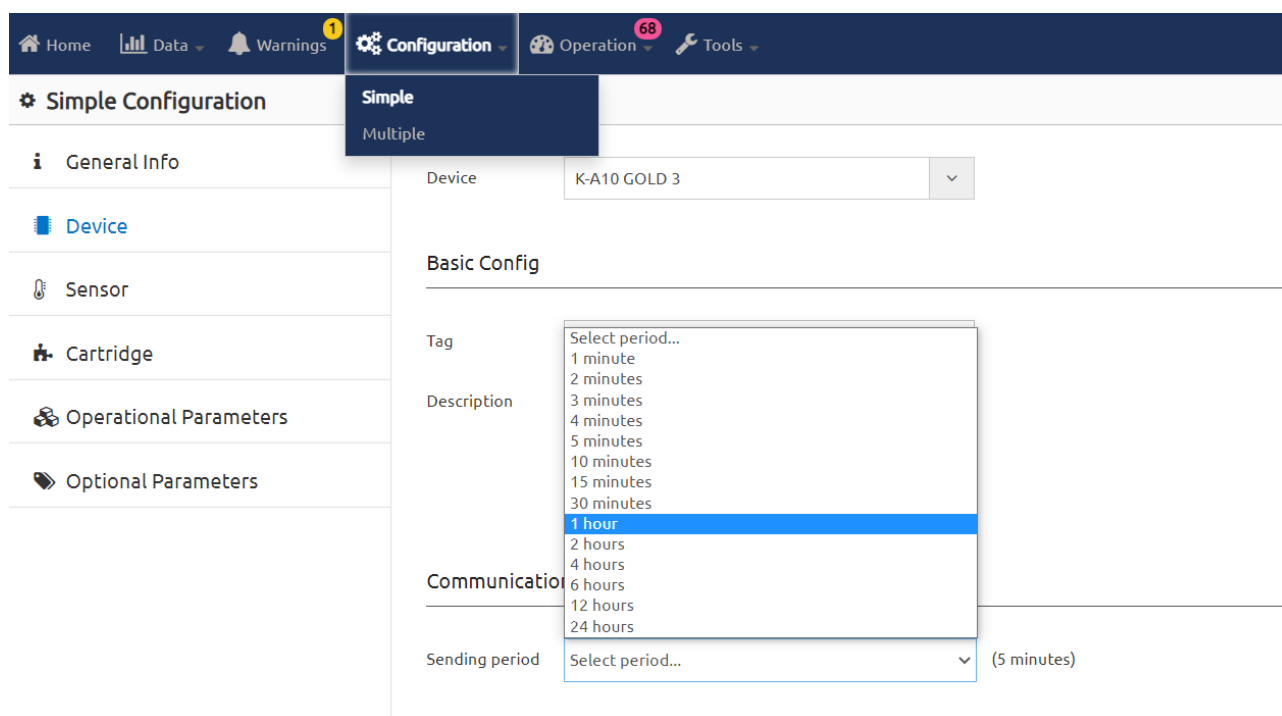


Figure 4 – Change of the sending period

- It is not recommended to use dry air (see Table 1 – Temperature, humidity and pressure condition for the test) for the calibration process. Please, note that the whole calibration process should be carried out with similar humidity conditions for a long time.

Two calibrations must be done to guarantee the correct performance of the **Kunak Air**:

- **Baseline:** This procedure sets the zero value of the sensors. It must always be done before the Span Calibration.
 1. Introduce zero air to the DUT and check that the humidity has not changed dramatically. If there is a big change in the humidity conditions, continue in this step for 4 hours.
 2. Check the concentration values with the reference equipment.
 3. Wait 8 hours, for sensors stabilization and check that no humidity transients are presented during this time. Also, check that the sensor readings are stable during the last 2 hours.
 4. Calibrate the sensor baseline using the **KunakCloud** web Calibration Tool.
 5. After calibration, wait 1 hour and confirm that the sensor value is zero. If not, repeat the previous step.



NOTE: The sensors accuracy at zero concentration is expected to be around ± 5 ppb for NO, NO₂, O₃, H₂S and SO₂ and ± 20 ppb for the CO if the steps have been correctly followed and no humidity transients are present during the calibration procedure

- **Span:** This procedure adjusts the sensitivity of the sensor.
 1. Introduce a constant concentration of the desired gas. Recommended concentrations are between 100 ppb and 400 ppb for NO, NO₂, O₃, H₂S, SO₂; between 1ppm to 8ppm for CO; and between 1 ppm and 10 ppm for VOCs. In the last case, and Isobutylene gas bottle is recommended for span calibration. It must be taken into account all the impurities that can be introduced in the enclosure, especially with the O₃ and NO_x sensors.
 2. Check the concentration of the gas with the reference equipment. Always use the reference equipment readings for the Span calibration set point.

3. Wait 1 hour for the sensor to stabilize.
4. Calibrate the sensor using the **KunakCloud** web Calibration Tool.
5. After calibration, wait 1 hour to confirm that the concentration is the same as the reference equipment.



NOTE: The baseline calibration must be done before the span calibration

1.1.2.3 Validation tests

Every **Kunak Air** has passed the “Kunak quality system assurance” before shipping. However, if any kind of test might be done at a reference laboratory, we recommend carrying out the tests below with the setup described in section 1.1.2.

- **Test A:** Introduce different gas concentrations and compare the results with the reference equipment. The typical concentration setpoints are shown in Figure 5 – Test of gas concentration. In the case of CO, it is recommended to do steps from 1 ppm to 8 ppm with a minimum interval of 1 hour between steps.

In the case of VOCs, an Isobutylene gas bottle must be used. In this case, it is recommended to do steps: 10 ppm, 5 ppm, 1 ppm and 0.5 ppm.

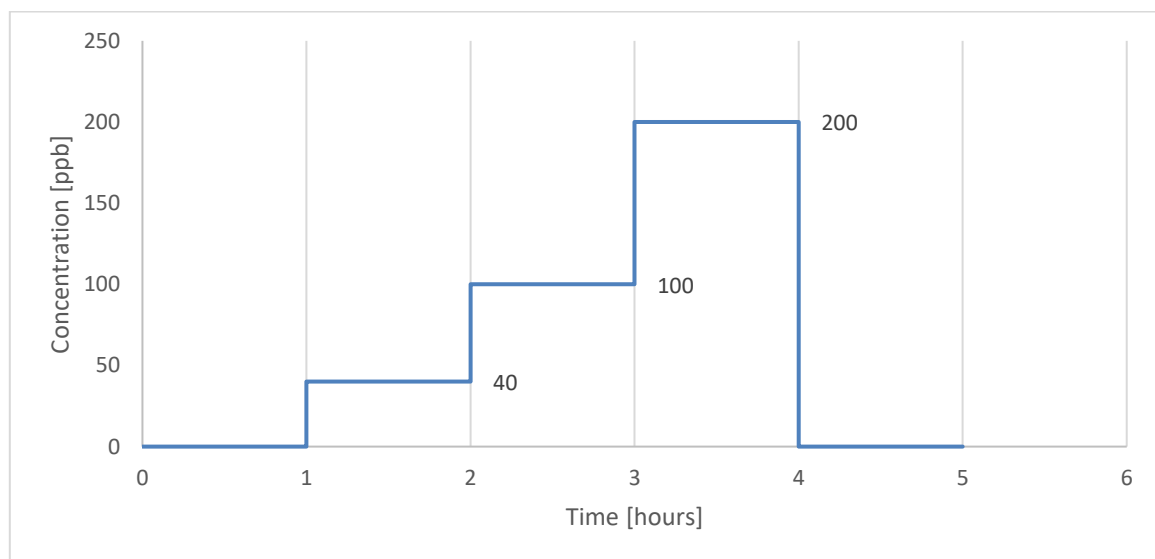


Figure 5 – Test of gas concentration

- **Test B:** With a constant concentration of one of the gases, change the temperature and the humidity and compare it with the reference equipment. Recommended to made progressive changes and not exceed ± 5 °C per hour and $\pm 10\%$ RH per hour.

Considerations:

- A stabilization of 4 hours will be required if abrupt changes in temperature or humidity occur.
- The equipment recalculates the concentration of the gas depending on the temperature and humidity values. That is why the sensors of humidity and temperature must be in contact with the measured gas.

1.1.3 Baseline calibration

This calibration consists of finding a period, or bringing the sensor, under a minimum reference gas concentration, ideally zero gas concentration of the targeted gas, and modify the calibration parameters under this condition.



NOTE: Please take into account that this calibration mustn't work correctly if it is applied to other values than the minimum gas concentrations, since it modifies the zero conditions of the sensors.

Gases will usually hit a minimum concentration during a certain part of the day. For example, it is possible to find concentrations that vary, depending on the type of gas, from 0 ppb to a few ppb at night time. The user must find these periods to perform a correct baseline calibration.

Follow the steps below to perform a correct baseline calibration through the **KunakCloud** web.

1.1.3.1 Calibration Type, Devices and Gas

The very first step in a calibration process is selecting the device(s) and gas to calibrate. It is possible to calibrate multiple devices at the same time by selecting them in the dropdown list or writing its tag. Note that only one gas can be calibrated at a time. To go to the next step, click on the "Next" button.

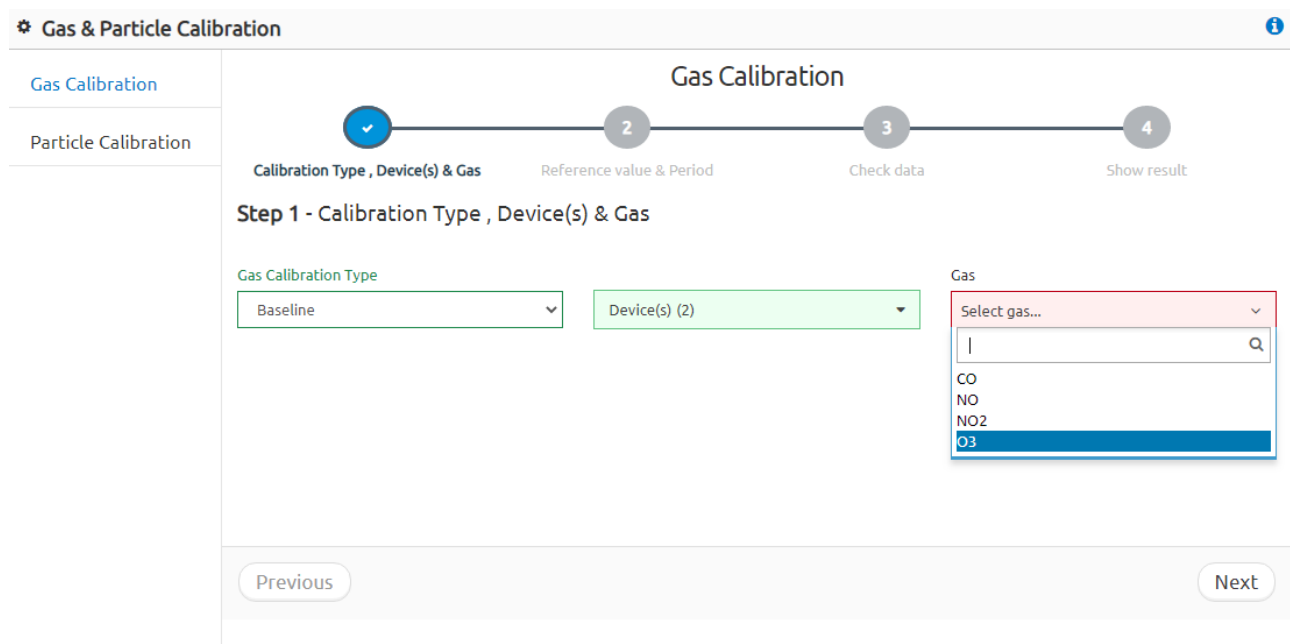


Figure 6 – Baseline calibration: select device & gas

1.1.3.2 Reference value and Period

Select the period in which the gas concentration has reached the reference value. In Figure 7 – Baseline calibration: calibration period selection, it is selected a reference value of 0 ppb starting at 1:00 am of January 21st 2021 for 2 hours.

Gas Calibration

Step 2 - Reference value & Period

Select the period within which the gas concentration reached the reference value

Reference value in ppb (min)

Start date

Start time

Duration

Time zone

Previous
Next

Figure 7 – Baseline calibration: calibration period selection

1.1.3.3 Check data

In this step, you must select the period to be charted so you can see the results of your calibration in the existent data before applying the calibration. Optionally, two reference devices with their corresponding gas can be selected.

Gas Calibration

Step 3 - Check data

Period to be charted and if required to rebuild

Start date

Start time

End date

End time

Reference device(s) i

Device 1:

Device 2:

Parameter 1:

Parameter 2:

Figure 8 – Baseline calibration: check data

1.1.3.4 Show result

Before applying the calibration, the chart specified in the previous step will be plotted showing how the calibration will be. The curves that will appear in the chart shown in Figure 9 – Baseline calibration: apply calibration are:

- **Blue** line (K-A14 Device 16-NO2 GC cal) is the calibrated concentration. The tag follows the format '{calibrated device name}-{gas name} cal'.
- **Red** line (K-A14 Device 16-NO2 GC) is the original concentration. The tag follows the format '{calibrated device name}-{gas name}'.
- **Green** line (RS Antwerpen ref-PM10 ref) is the reference sensor. Up to two reference sensors can be selected and they will be tagged as '{reference device name}-{sensor name} ref'.

The statistics of calibration against reference are calculated with the original data and with the calibrated data. It shows the R2, the Minimum Absolute Error (MAE), the expanded uncertainty (U(exp)%) and the values above the Limit Value set by the Data Quality Objectives of the European Directive 2008/50/EC.

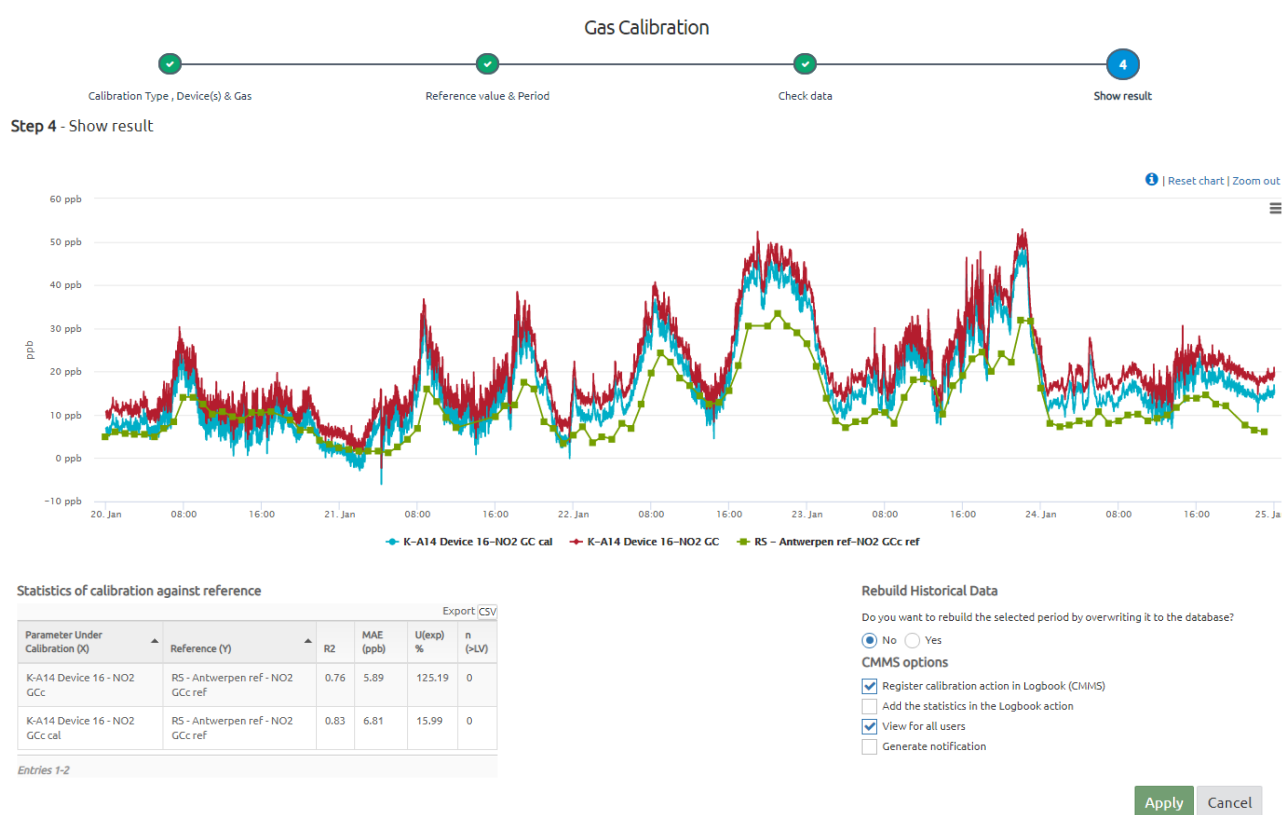


Figure 9 – Baseline calibration: apply calibration

If the values of the chart are correct, it exists the possibility to “Rebuild Historical data”, as appeared in Figure 10 – “Rebuild historical data” option, by overwriting the data with the new calibrated values over the selected period by clicking on the “Yes” option when asking to rebuild the data. Besides, you have the possibility to register the calibration in the CMMS:

- Register calibration action in the Logbook (CMMS): it will appear in the Logbook, the device calibrated, the person who carried out the calibration, the description of the calibration (for instance, “Baseline calibration – NO2”), and the time period for which the calibration was applied.
- View for all users: check this option if you want that all the users that have the device linked, can see the calibration done.
- Generate notification: it will generate a notification to other users that calibration was performed.

Rebuild Historical Data

Do you want to rebuild the selected period by overwriting it to the database?

☒ No ☐ Yes

CMMS options

☒ Register calibration action in Logbook (CMMS)

☒ View for all users

☐ Generate notification

Apply Cancel

Figure 10 – “Rebuild historical data” option

Finally, by clicking on the “Apply” button a confirmation dialog message will be shown. Clicking on “YES”, a completed message will appear at the top-right of the web. The new settings will be queued and applied to start from the next communication period.

In case that the preliminary result of the calibration is not as expected, just click on “Cancel” button to abort the calibration or click on “Previous” to modify the settings.

1.1.3.5 Example

Before calibrating a gas, we need to study its behavior to detect a specific period where we know how it should behave.

Let us suppose we want to calibrate the NO₂ concentration of an air quality device tagged as 'K-A14 Device 16'. This device has the following NO₂ historical data (Data > Standard):

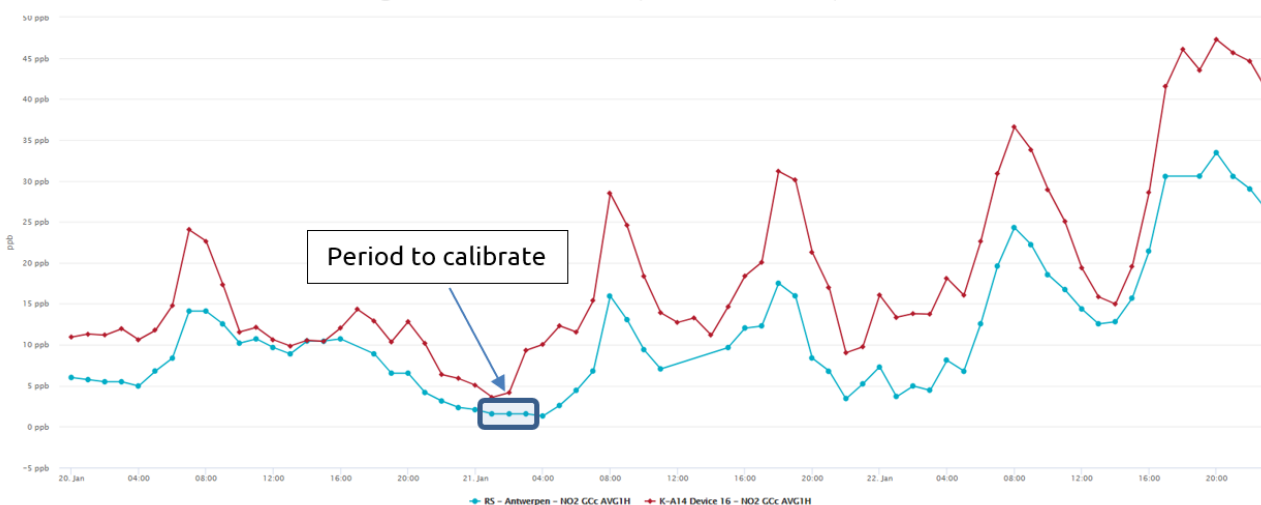


Figure 11 – Baseline calibration: preliminary concentration analysis

We are going to use another device as a reference ('RS - Antwerpen '). This device should be placed in the same location as the one we want to calibrate.

As it is known, NO₂ tends to be zero during the night in the location where both devices are located. In this case, the period between 01:00 and 03:00 should show a mean concentration of zero.

Once the preliminary concentration analysis is performed, you can start with the baseline calibration from “Operation” tab > “Gas & Particle Calibration” > “Gas calibration”.

The first step is to select the 'K-A14 Device 16" device and the NO2 sensor in the multiple device and single gas selectors:

Gas Calibration

1 — 2 — 3 — 4

Calibration Type , Device(s) & Gas
Reference value & Period
Check data
Show result

Step 1 - Calibration Type , Device(s) & Gas

Gas Calibration Type

Baseline

Device(s) (1)

All
Selected

device 16

Select all
Deselect all

☒
K-A14 Device 16

Gas

NO2

Previous

Next

Figure 12 – Baseline calibration example: select device and gas

Since the NO2 concentration should be zero from 01:00 to 03:00, we will select a period of 2 hours, starting the 2021-01-21 at 02:00 am in step 2:

Gas Calibration

✓ — ✓ — 3 — 4

Calibration Type , Device(s) & Gas
Reference value & Period
Check data
Show result

Step 2 - Reference value & Period

Select the period within which the gas concentration reached the reference value

Reference value in ppb (min)

0

Start date

2021-01-21

📅

Start time

01:00

🕒

Duration

2 hours

Time zone

(UTC+1:00) Amsterdam, Berlin, Vienna, Rome, B

Previous

Next

Figure 13 – Baseline calibration example: select calibration period

Once the calibration period has been selected, in step 3 you must select the period to be analysed. This period is a simulation of the calibration applied to already stored concentration values:

Gas Calibration

✓ — ✓ — 3 — 4

Calibration Type , Device(s) & Gas Reference value & Period **Check data** Show result

Step 3 - Check data

Period to be charted and if required to rebuild

Start date

2021-01-20
📅

End date

2021-01-24
📅

Start time

00:00
🕒

End time

23:59
🕒

Reference device(s) ❗

Device 1:

RS - Antwerpen
▼

Device 2:

Select device...
▼

Parameter 1:

NO2 GCc
▼

Parameter 2:

Select parameter...
▼

Figure 14 – Baseline calibration example: check data

The mean of the calibrated concentration during the selected calibration must be 0, and this can be checked in step 4:

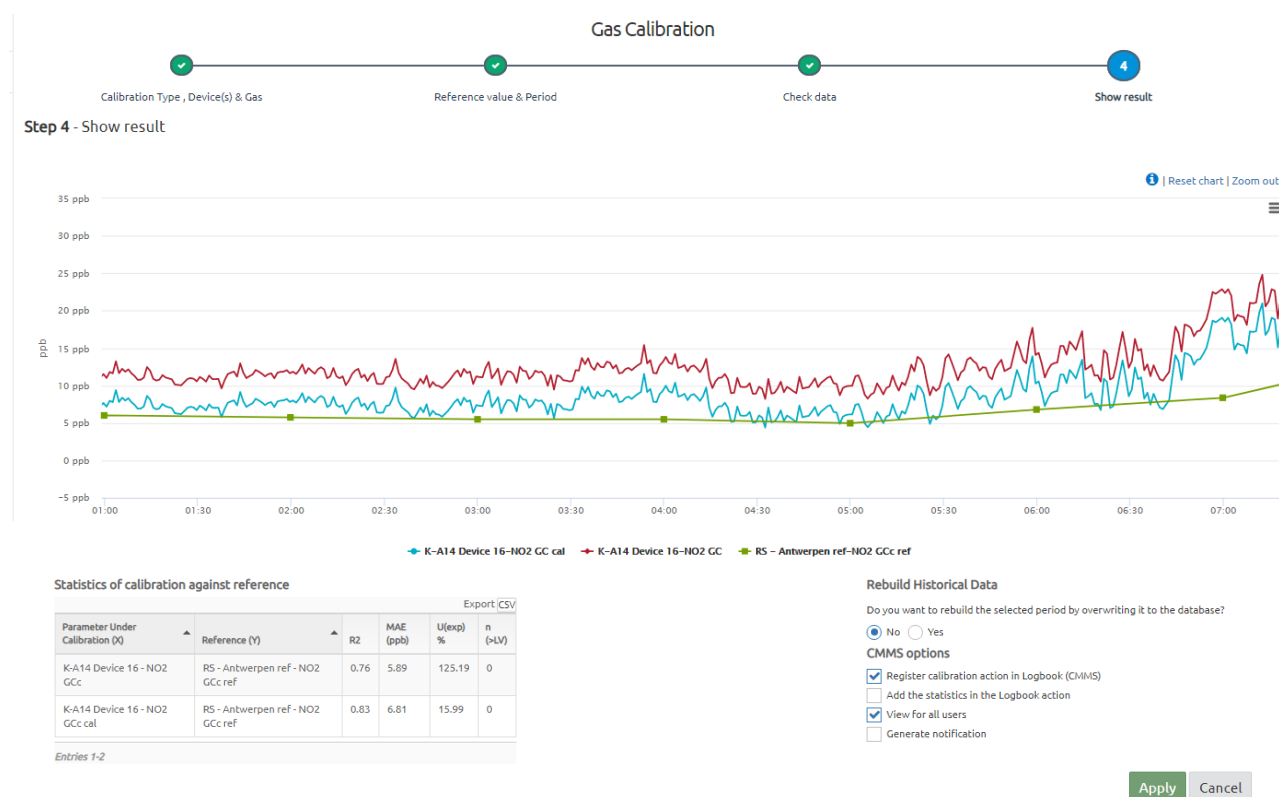


Figure 15 – Baseline calibration example: average concentration within the calibration period

Applying a zoom on the desired period, the mean is 0, so the calibration has worked as expected. Now let's see the effects of the calibration on the original curve:

Applying zoom out on the whole selected period, the curve offset has decreased, and it is now quite similar to the reference curve of the 'RS – Antwerpen' device:

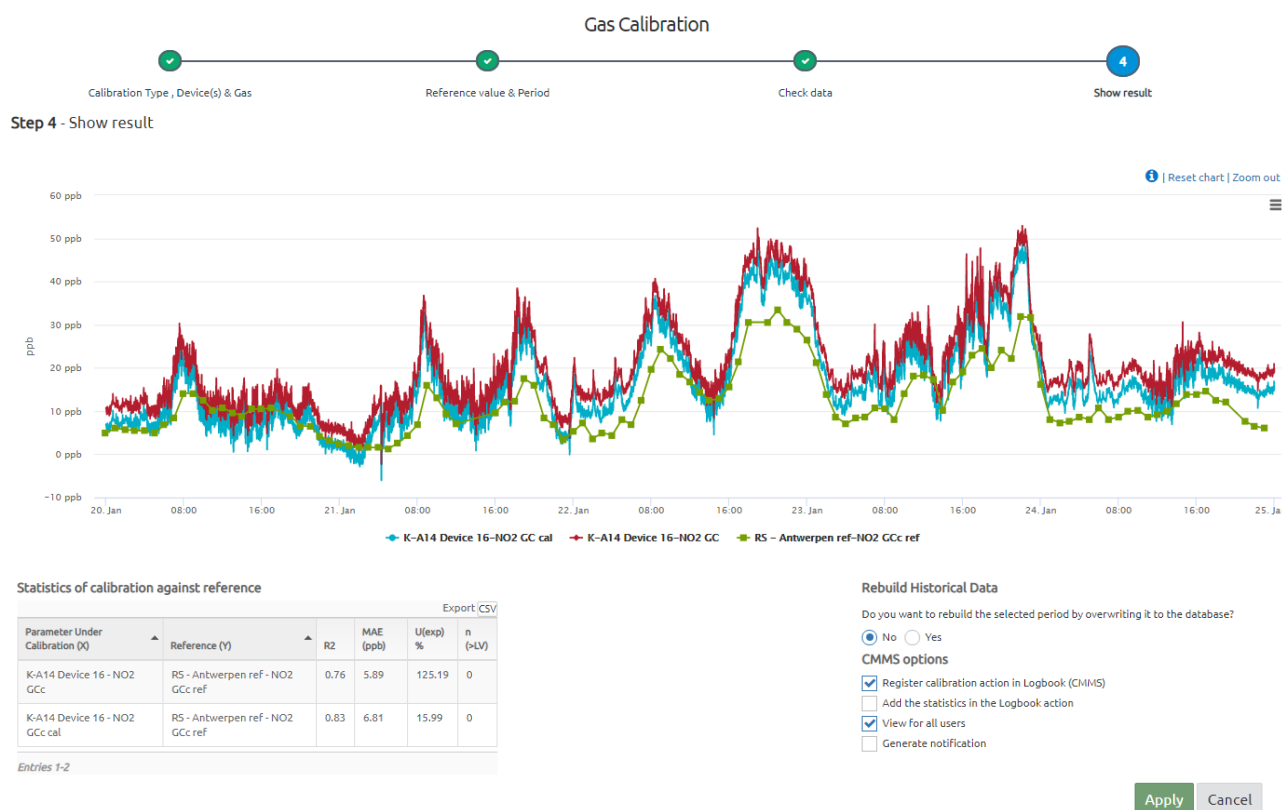


Figure 16 – Baseline calibration example: calibrated curve vs reference curve

The calibration seems to be correct so now we can “Rebuild the Historical Data” by clicking on the “Yes” option and select as required the different “CMMS options”. Finally, apply the calibration by clicking on the “Apply” button.

After confirming the calibration, the new settings will be queued and applied to start from the next communication period.

1.1.4 Span calibration

This calibration consists of adjusting the sensor's span by finding a period in which the gas concentration reached a maximum reference value, or failing that, a very high value. It modifies the sensitivity of the air quality sensors so the concentration span will change accordingly. The span value will be expanded upon setting a maximum value higher than current concentrations, while it will be constrained upon setting a maximum value lower than current concentrations. It is very important to note that this calibration will not work correctly if the values applied are not the maximum or the highest values concentrations, since it modifies the sensitivity parameter of the gas sensors.



NOTE: Please take into account that a previous baseline calibration must be done or be sure that the baseline is correct.

Gases will hit a maximum concentration, or peaks of concentration, during a certain part of the day. For example, it is possible to find high gas concentrations in the city center during the early morning hours due to the dense car traffic. The user must find these periods to perform a correct span calibration when compared with a GOLD unit or reference instrument.

To calibrate the span, it is recommended to take high concentration values as a reference. The recommended minimum values are shown in Table 2 – Minimum recommended values for calibrating the span.

Parameter	Minimum values recommended
Carbon monoxide (CO)	700 ppb
Carbon dioxide (CO ₂)	1000 ppm
Ozone (O ₃)	40 ppb
Nitric oxide (NO)	40 ppb
Nitrogen dioxide (NO ₂)	40 ppb
Hydrogen sulphide (H ₂ S)	40 ppb
Sulphur dioxide (SO ₂)	40 ppb
Ammonia (NH ₃)	20 ppm
Volatile Organic Compounds (VOCs)	5 ppm

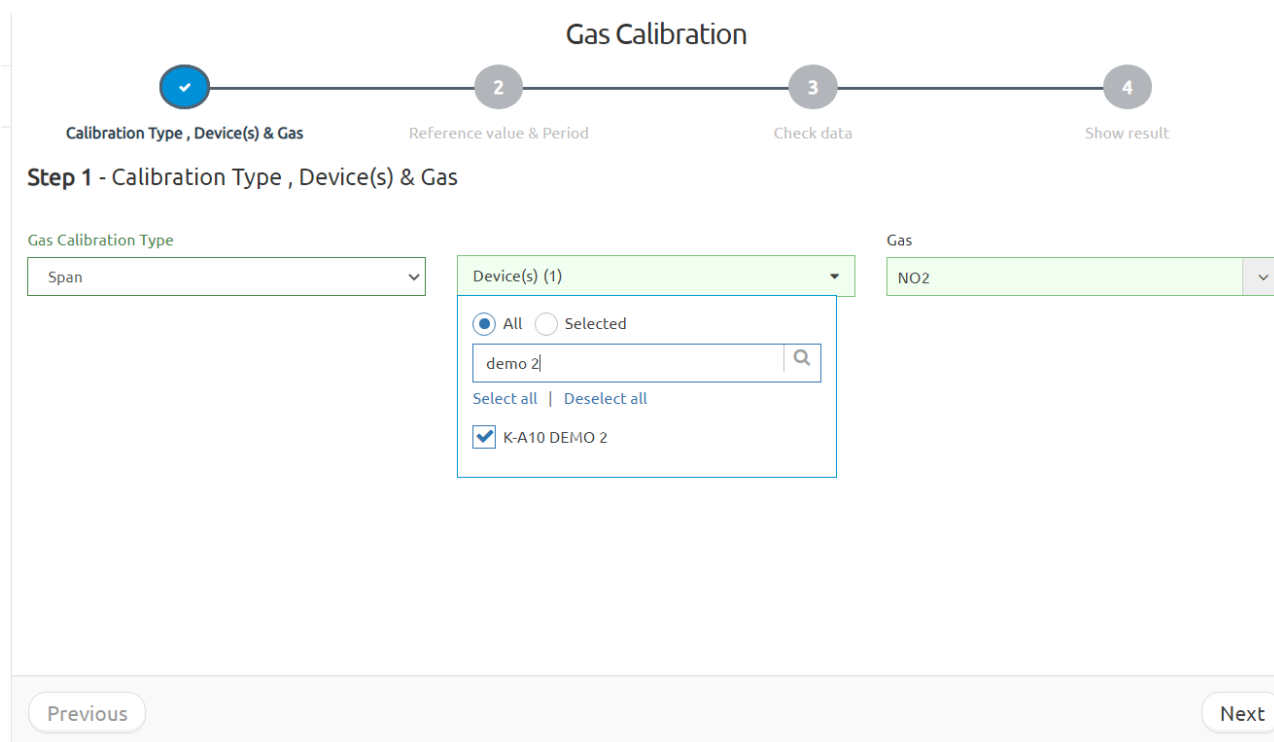
Table 2 – Minimum recommended values for calibrating the span

Follow the steps below to perform a correct span calibration through the **KunakCloud** web.

NOTE: The gas considered for VOCs cartridge span calibration is the Isobutylene

1.1.4.1 Calibration Type, Devices and Gas

The very first step in a calibration process is selecting the device(s) and gas to calibrate. It is possible to calibrate multiple devices at the same time by selecting them in the dropdown list or writing its tag. Note that only one gas can be calibrated at a time. To go to the next step, click on the “Next” button.



Gas Calibration

1 2 3 4
 Calibration Type , Device(s) & Gas Reference value & Period Check data Show result

Step 1 - Calibration Type , Device(s) & Gas

Gas Calibration Type: Span

Device(s) (1): demo 2

Gas: NO2

☒ All ☐ Selected
 Select all | Deselect all
☒ K-A10 DEMO 2

Previous Next

Figure 17 – Span calibration: select device & gas



NOTE: The span calibration of the carbon monoxide **Type B** cartridge must be done at laboratory conditions by using a CO gas bottle. Kunak recommend carrying out the CO Type B span calibration at 100 ppm of CO.

1.1.4.2 Reference value and Period

Span calibration can only be done when comparing to a GOLD unit or a reference instrument collocated with the device under calibration.

Select the period in which the gas concentration reached the reference value. In Figure 18 – Span calibration: calibration period selection, it is selected a reference value of 51 ppb starting at 19:00 of July 22nd and for 1 hour.

Gas Calibration

1
Calibration Type , Device(s) & Gas

2
Reference value & Period

3
Check data

4
Show result

Step 2 - Reference value & Period

Select the period within which the gas concentration reached the reference value

Reference value in ppb (max)

51

Start date

2020-07-22

📅

Start time

19:00

🕒

Duration

1 hour

▼

Time zone

(UTC+1:00) Brussels, Copenhagen, Madrid, Paris

▼

Previous

Next

Figure 18 – Span calibration: calibration period selection

In span calibration, the selected period should be a single point or a few points (when the value is constant or almost constant during a particular period). The idea is to expand or collapse the curve span (the difference between the maximum and minimum value of the concentration) by adjusting its value. The best way is to use the average hour to calibrate the span precisely.

1.1.4.3 Check data

In this step, you must select the period to be charted so you can see the results of your calibration in the existent data. Optionally, two reference devices with their corresponding gas can be selected.

Gas Calibration

Step 3 - Check data

Period to be charted and if required to rebuild

Start date 2020-07-22	Start time 00:00
End date 2020-07-22	End time 23:59

Reference device(s)

Device 1: K-A10 GOLD 3	Parameter 1: NO2 GCc
Device 2: Select device...	Parameter 2: Select parameter...

Previous
Next

Figure 19 – Span calibration: check data

1.1.4.4 Show result

Before applying the calibration, the chart specified in the previous step will be plotted showing how the calibration will be. The curves that will appear in the chart shown in Figure 20 – Span calibration: apply calibration are:

- **Red** line (K-A10 Demo 2 NO2 GC) is the original concentration. The tag follows the format '{calibrated device name}-{gas name}'.
- **Blue** line (K-A10 Demo 2-NO2 GC cal) is the calibrated concentration. The tag follows the format '{calibrated device name}-{gas name} cal'.
- **Green** line (K-A10 Gold 3-NO2 GCc ref) is the optional reference sensor. Up to two reference sensors can be selected and they will be tagged as '{reference device name}-{sensor name} ref'.

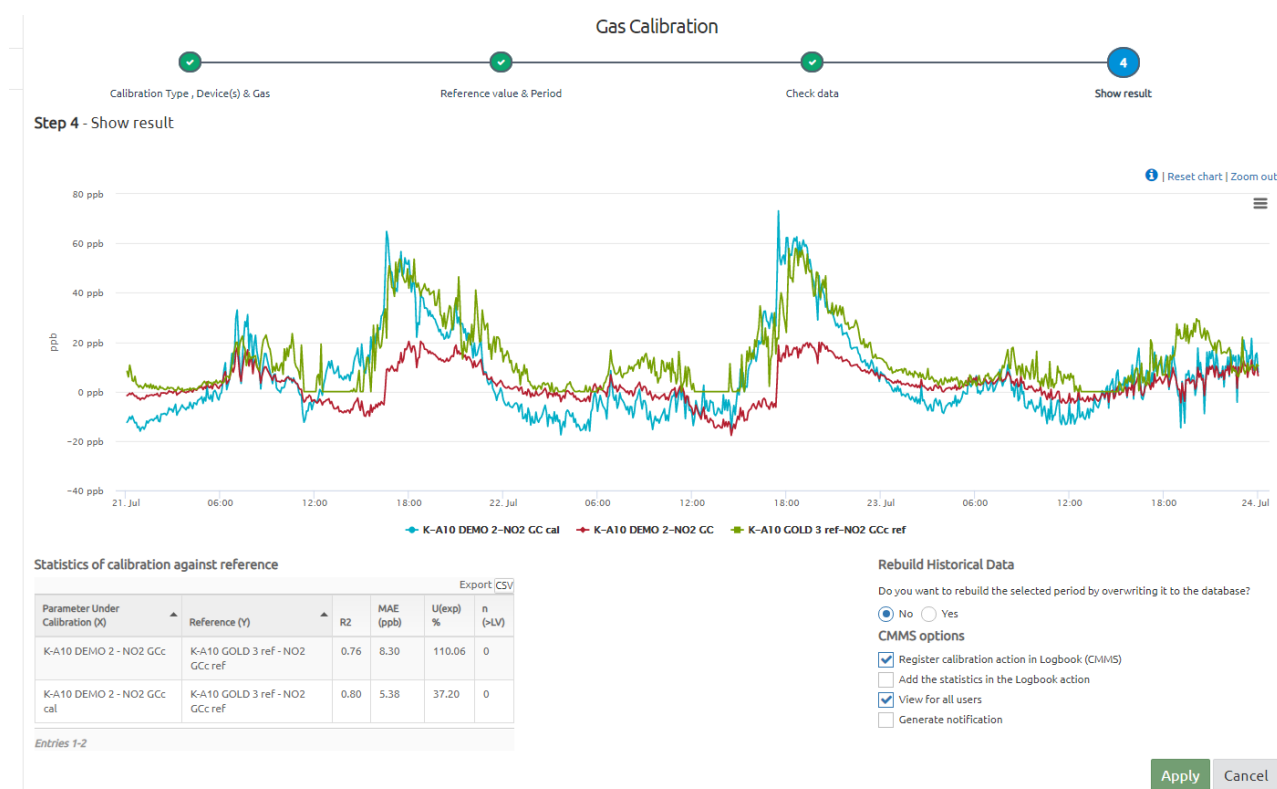


Figure 20 – Span calibration: apply calibration

If the values of the chart are correct, it exists the possibility to “Rebuild Historical data”, as appeared in Figure 10 – “Rebuild historical data” option, by overwriting the data with the new calibrated values over the selected period by clicking on the “Yes” option when asking to rebuild the data. Besides, you have the possibility to register the calibration in the CMMS. For more details see the *Baseline calibration* Section.

Finally, by clicking on the “Apply” button a confirmation dialog message will be shown. Clicking on “YES”, a completed message will appear at the top-right of the web. The new settings will be queued and applied to start from the next communication period.

In the case that the preliminary result of the calibration is not as expected, just click on “Cancel” button to abort the calibration or click on “previous” to modify the settings.

1.1.4.5 Example

It is considered in this example that the calibration is already done in [the example of the baseline calibration](#) before and it is queued to the device.

You must always be sure that the baseline calibration has been done.

This span calibration will adjust the peaks of the concentration, so the calibrated concentration will be even more similar to the reference concentration.

The NO₂ concentration of the air quality device tagged as 'K-A10 DEMO 2' was the following one:

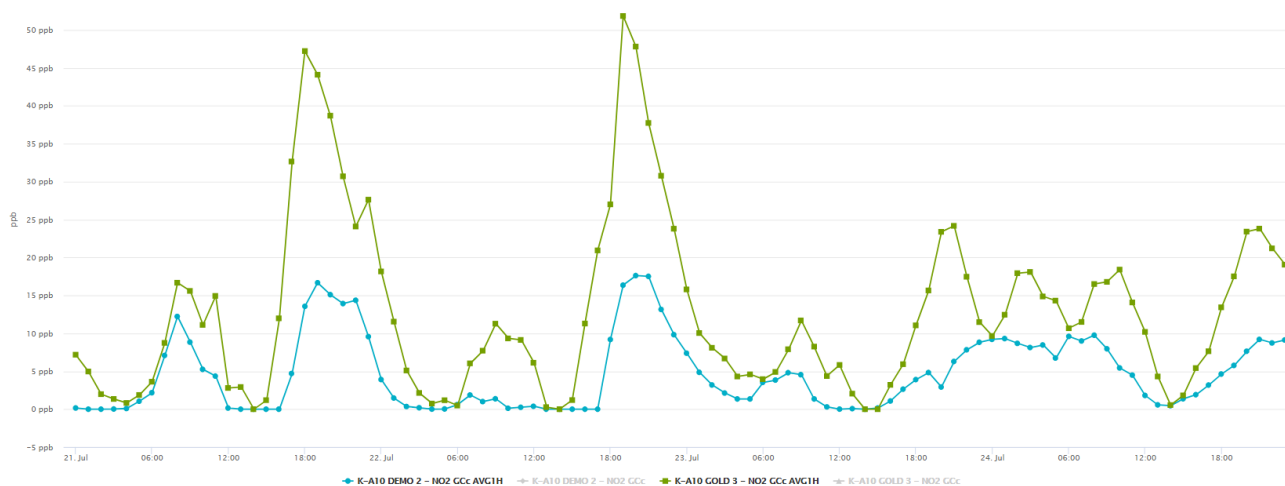


Figure 21 – Span calibration: previous concentration analysis

There was a good correlation with the concentration of the reference device (K-A10 GOLD 3). The curves are quite similar but there are still some peaks in the calibrated device that could be adjusted. We will focus attention on the interval with a peak at 19:00 from July 22nd 2020:

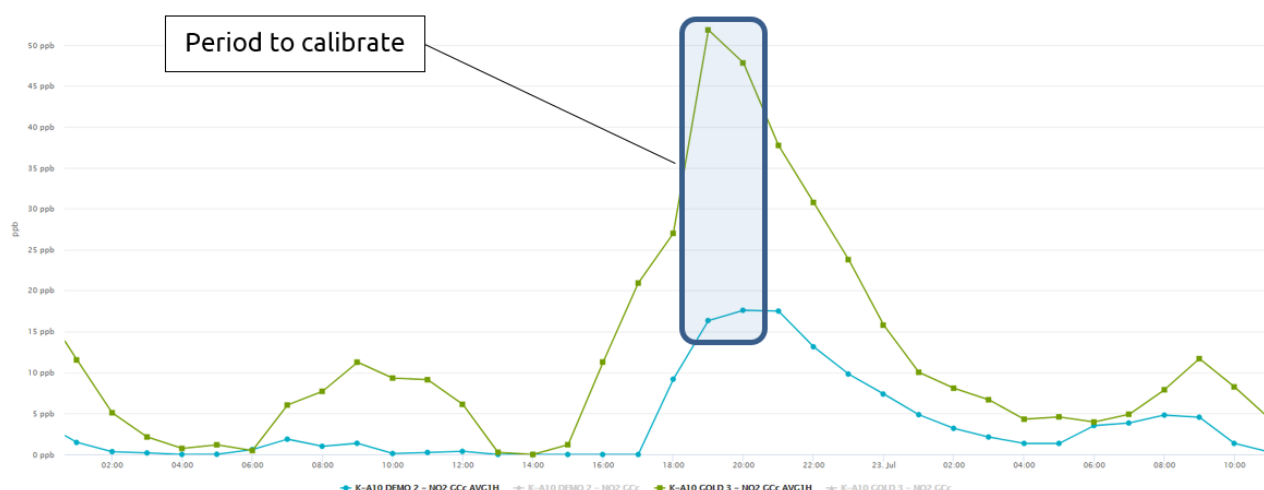


Figure 22 – Span calibration example: select the reference period

It is better to calibrate the span taking into account the average hour because the error made during calibration will be smaller.

We will try to adjust the span to the average NO₂ concentration of the reference device. By using the average concentration as a reference, the influence of the extreme values (abnormal peaks) is decreased. The average NO₂ concentration between 19:00 and 20:00 is 51 ppb for the reference and the "K-A10 DEMO 2" is 17 ppb as it is shown in Figure 22 – Span calibration example: select the reference period.

Once the reference value (51 ppb) has been found, the 'K-A10 DEMO 2" and the NO₂ sensor must be selected at the calibration tool.

Gas Calibration

1
Calibration Type , Device(s) & Gas

2
Reference value & Period

3
Check data

4
Show result

Step 1 - Calibration Type , Device(s) & Gas

Gas Calibration Type

Span

Device(s) (1)

All
Selected

demo 2

Select all
Deselect all

☒ K-A10 DEMO 2

Gas

NO2

Previous

Next

Figure 23 – Span calibration example: select device and gas

We will select 51 as a reference and a period from 19:00 to 20:00 from July 22nd, 2020:

Gas Calibration

1
Calibration Type , Device(s) & Gas

2
Reference value & Period

3
Check data

4
Show result

Step 2 - Reference value & Period

Select the period within which the gas concentration reached the reference value

Reference value in ppb (max)

51

Start date

2020-07-22

Start time

19:00

Duration

1 hour

Time zone

(UTC+1:00) Brussels, Copenhagen, Madrid, Paris

Previous

Next

Figure 24 – Span calibration example: select calibration period

As a reference, we select the 'NO2 GCc' sensors of the reference device (K-A10 GOLD 3) to check if the results are as expected:

Gas Calibration

✓ ✓ 3 4

Calibration Type , Device(s) & Gas
Reference value & Period
Check data
Show result

Step 3 - Check data

Period to be charted and if required to rebuild

Start date

2020-07-22

Start time

00:00

End date

2020-07-22

End time

23:59

Reference device(s) i

Device 1:

K-A10 GOLD 3

Parameter 1:

NO2 GCc

Device 2:

Select device...

Parameter 2:

Select parameter...

Previous
Next

Figure 25 – Span calibration example: show result

In this last step, we will check if everything worked as expected:



Figure 26 – Span calibration example: apply calibration

The curve has been collapsed and adjusted span. Now, we can compare this calibrated concentration with the reference one:



Figure 27 – Span calibration example: the result of combining baseline and span calibration

The calibration seems to be correct so now we can “Rebuild the Historical Data” by clicking on the “Yes” option and select as required the different “CMMS options”. Finally, apply the calibration by clicking on the “Apply” button.

After confirming the calibration, the new settings will be queued and applied to start from the next communication period.

1.2 Particle calibration

Correction applied to the sensor data is particularly important in the case of particle measurements. The state-of-the-art of PM sensing technologies applicable in compact sensors measures the particle mass density with indirect means, which are commonly based on the scattering of light from atmospheric particles. In these methods, variations in particle density, shape and optical properties inevitably create biases in the particle mass measurement. These biases can be compensated by comparison with a reference, or reference equivalent method, after typically linear correction factors (slope/intercept) are established for adjusting the sensor data. That difference or variation could be higher depending on the method used to measure in the reference instrument.

The correction factors can be stored and applied to the data at various stages of data acquisition: in the instrument itself, at a network database, or as a user applied post-processing step. It should be taken care to keep track of correction factors to avoid errors caused by applying the factors multiple times to the same data.

To apply this correction factor in the instrument itself, you can use our **KunakCloud** platform and do it remotely through “Particle Calibration” tool that can be found in “Operation” tab > “Gas & Particle Calibration” in the **KunakCloud** section.

This calibration factor can be applied in two ways: “Manual” or “Linear regression (forced origin) against reference”.

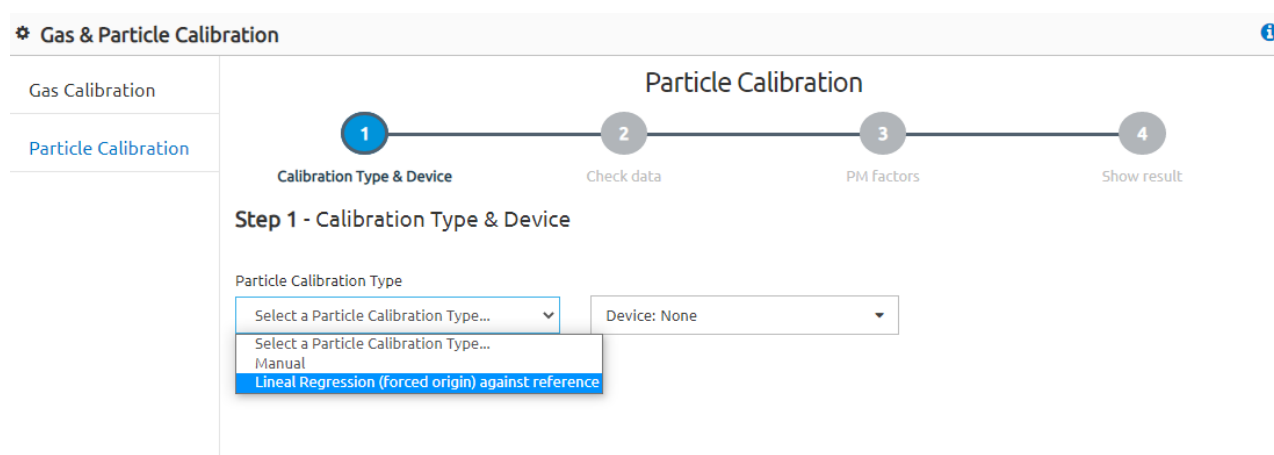


Figure 28 – Particle Calibration tool

For both particle calibration modes, place the **Kunak Air** device near an official reference station to carry out a colocation test. For the most accurate calibration, place the **Kunak Air** at a maximum distance of 2 meters from the inlet port of the reference air quality station.

The reference data used can be either a Reference Station or a GOLD device, as it occurs for gas calibrations.

The Kunak Air should be installed for at least 1 week to compare the **Kunak Air** PM data obtained against the reference data. After calibration, keep on the device working for at least another week to compare the data with the reference station or the GOLD device in order to guarantee a correct performance as specified in the datasheet and/or shown in the field tests that can be found at the “Support” section of the **KunakCloud** web.

Once the calibration has been validated, switch off the unit. Now, **Kunak Air** is ready to be deployed anywhere. If required, Kunak can integrate² into **KunakCloud** platform any third-party data (for example, air quality reference stations data) to make easier the calibration process.

In case you used a GOLD device, once the calibration has been validated, switch off the GOLD unit. Now, the GOLD device can be moved to another location to calibrate another device.

1.2.1 Lineal Regression (forced origin) against reference

The “Lineal Regression (forced origin) against reference” will calculate the linear regression between the reference instrument and the one that needs to be corrected. It will calculate the slope through the origin that will apply to the data.

Follow the steps below to perform a correct “Lineal Regression (forced origin) against reference” particle calibration through the **KunakCloud** web.

1.2.1.1 Calibration Type and Device

The very first step in a calibration process is the type of Particle Calibration, in this case, “Lineal Regression (forced origin) against reference” and then select the device to calibrate. To go to the next step, click on the “Next” button.

As an example, we use the device K-A14 Device 15.

² The data of the reference station to integrate in the KunakCloud must be public and correctly posted on the internet or a place that it can be accessible to Kunak Software to collect the data.

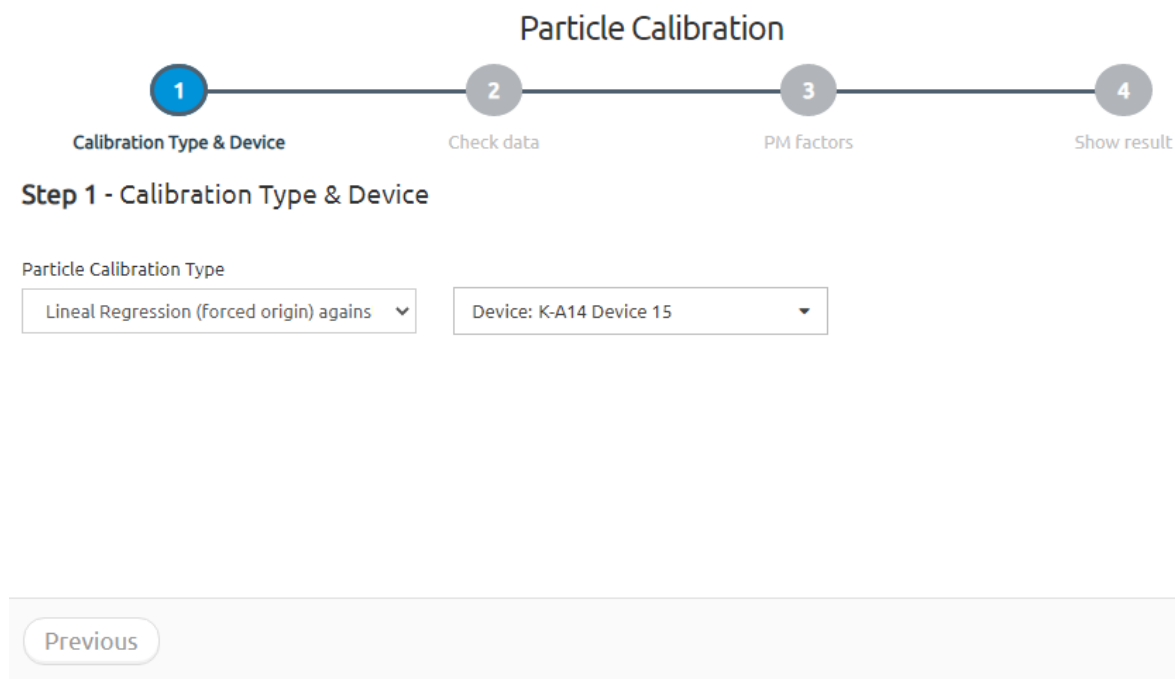


Figure 29 – Lineal Regression (forced origin) against reference calibration tool.

1.2.1.2 Check data

In this step, you must select the period to be charted and if required to rebuild, so the PM factors used for the calibration will be calculated and then, the results of your calibration in the existent data before applying the calibration will be shown. Optionally, it exists 3 reference devices to select, one for each particle size (PM₁₀, PM_{2.5} and PM₁).

As an example, we choose the dates from 27th November 2020 until 4th of December 2020, and the RS – Antwerpen as reference station for the PM₁₀, PM_{2.5} and PM₁. Thus, we select the RS – Antwerpen in the “Reference Device with PM10” selecting box, and then, “PM10” as “Parameter PM10”. The same steps are done for PM_{2.5} and PM₁.

To go to the next step, click on the “Next” button.

Particle Calibration

✓
 Calibration Type & Device

2
Check data

3
 PM Factors

4
 Show result

Step 2 - Check data

Period to be charted and if required to rebuild

Time zone
 (UTC+1:00) Brussels, Copenhagen, Madrid, Paris

Start date: 2020-11-28 Start time: 00:00

End date: 2020-12-04 End time: 23:59

Reference device(s) i

Reference Device with PM10 RS - Antwerpen	Parameter PM10: PM10
Reference Device with PM2.5 RS - Antwerpen	Parameter PM2.5: PM2.5
Reference Device with PM1 RS - Antwerpen	Parameter PM1: PM1

Previous
Next

Figure 30 – Particle calibration: check data

1.2.1.3 PM Factors

The new PM factors are calculated regarding the period selected in the step before. The “New factor” calculated in the slope forced to the origin, thus the “New offset” value will remain as 0.

In the first two columns the “Current factor” and “Current offset” values appear in case another calibration was performed before. Thus, if not prior calibration was carried out, the “Current factor” and “Current offset” will be 1 and 0 respectively, as shown in Figure 31 – Particle calibration: PM factors (not previous calibration). If a calibration has been performed previously, the “Current factor” and “Current offset” will show the values applied in the last calibration, as shown in Figure 32 – Particle calibration: PM factors (previous calibration performed). In the last case, the “New factor” will be calculated with the raw PM data, not with the calibrated data. The “Current factor”(s) will be deleted, and substitute by the “New factor”(s) calculated.

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1	0	0.9294006	0
PM2.5	1	0	1.0011568	0
PM1	1	0	2.3711138	0

Previous
Next

Figure 31 – Particle calibration: PM factors (not previous calibration)

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1.429646	0	1.3192322	0
PM2.5	1.296212	0	1.2070069	0
PM1	1.296212	0	2.576908	0

Previous
Next

Figure 32 – Particle calibration: PM factors (previous calibration performed)

A constraint appears if PM10 factor is lower than PM2.5 and PM1, and if PM2.5 factor is lower than PM1. The PM10 factor must be higher than PM2.5 and PM1 factors, as well as PM2.5 factor must be higher than PM1 factor. This constraint is added in order to avoid that PM1 and PM2.5 concentration values are higher than PM10 values after the calibration.

If the “New factor”(s) calculated shows that constraint when applying them, we recommend setting the same “New factor” obtained for PM10, for the case of PM2.5 and PM1. The PM calibration tool allows changing the PM “New factor”(s) calculated.

In Figure 33 – PM calibration: factor constraint PM10 factor > PM2.5 factor > PM1 factor appears an example. “New factor”(s) are calculated for PM10, PM2.5 and PM1. The constraint notification appears since PM1factor is higher than PM10 and PM2.5. This, it is recommended to change the PM1 factor as the same value as PM2.5 factor, 1.2070069, as shown in Figure 34 – PM calibration: factor constraint corrected.

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1.429646	0	1.3192322	0
			Parameters of PM10 are less than those of PM1 and PM2.5	Parameters of PM10 are less than those of PM1 and PM2.5
PM2.5	1.296212	0	1.2070069	0
			Parameters of PM2.5 are less than those of PM1 or greater than those of PM10	Parameters of PM2.5 are less than those of PM1 or greater than those of PM10
PM1	1.296212	0	2.576908	0
			Parameters of PM10 are greater than those of PM1 and PM2.5	Parameters of PM10 are greater than those of PM1 and PM2.5

Previous
Next

Figure 33 – PM calibration: factor constraint PM10 factor > PM2.5 factor > PM1 factor

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1.429646	0	1.3192322	0
PM2.5	1.296212	0	1.2070069	0
PM1	1.296212	0	1.2070069	0

Previous
Next

Figure 34 – PM calibration: factor constraint corrected

To select the “New factors”(s) click on “Next” button.

1.2.1.4 Show result

Before applying the calibration, the chart specified in the previous step will be plotted showing how the calibration will be. The curves that will appear in the chart shown are:

- **Blue** line (K-A14 Device 15 – PM10 cal) is the calibrated concentration. The tag follows the format '{calibrated device name}-{particle name} cal'.
- **Red** line (K-A14 Device 15 – PM10) is the original concentration. The tag follows the format '{calibrated device name}-{gas name}'.
- **Green** line (RS Antwerpen ref-PM10 ref) is the reference sensor. Up to three reference sensors, one for each particle size, can be selected and they will be tagged as '{reference device name}-{ sensor name} ref'.

The statistics of calibration against reference are calculated with the original data and with the calibrated data. It shows the R2, the Minimum Absolute Error (MAE), the expanded uncertainty (U(exp)%) and the values above the Limit Value set by the Data Quality Objectives of the European Directive 2008/50/EC.

If the values of the chart are correct, it exists the possibility to “Rebuild Historical data”, as appeared in by overwriting the data with the new calibrated values over the selected period by clicking on the “Yes” option when asking to rebuild the data. Only in the Particle Calibration tool, this option gives you the option to select the start and end date that you would like to rebuild data, as shown in Figure 36 – Rebuild Historical data: Dates setting. Besides, you have the possibility to register the calibration in the CMMS. For more details see the *Baseline calibration* Section.

Finally, by clicking on the “Apply” button a confirmation dialog message will be shown. Clicking on “YES”, a completed message will appear at the top-right of the web. The new settings will be queued and applied to start from the next communication period.

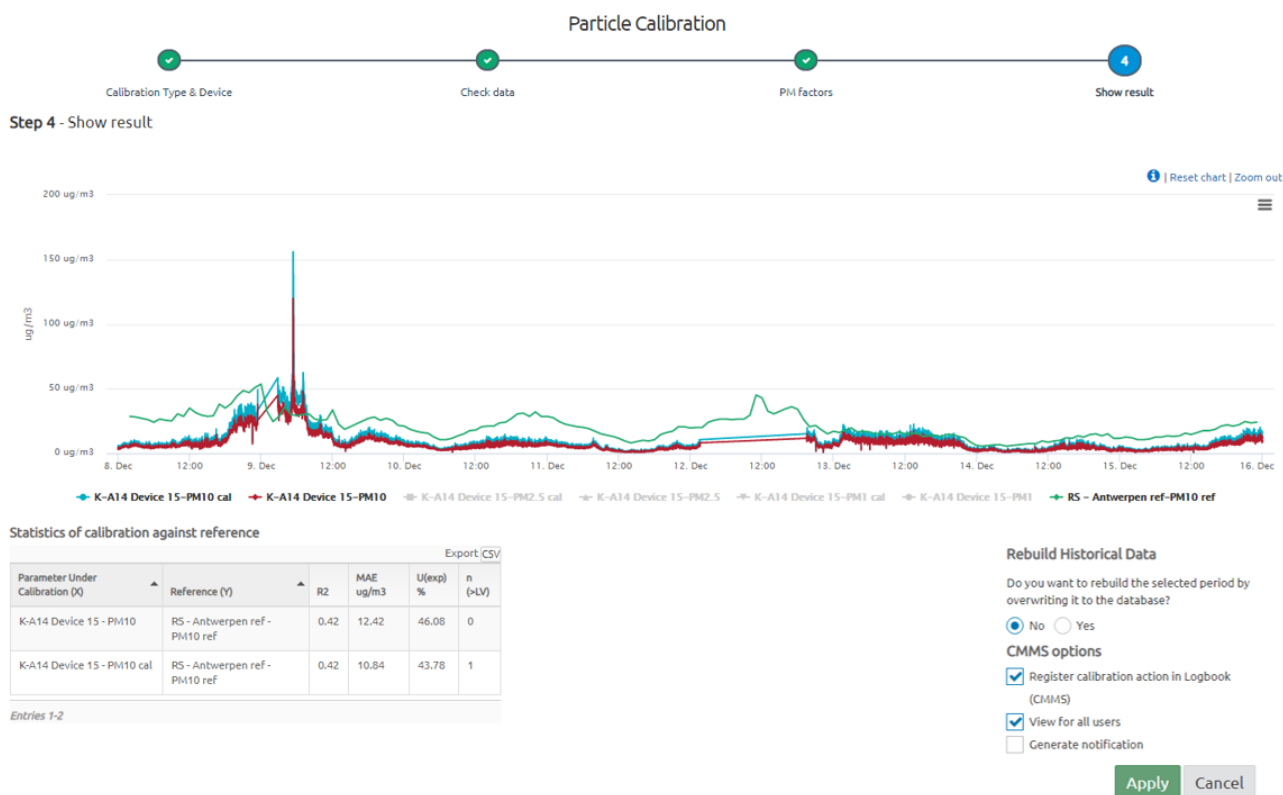


Figure 35 – Particle calibration: show results

Rebuild Historical Data

Do you want to rebuild the selected period by overwriting it to the database?

☐ No ☒ Yes

Start date

2021-01-31



Start time

00:00



End date

2021-02-04



End time

23:59



CMMS options

☒ Register calibration action in Logbook (CMMS)

☐ Add the statistics in the Logbook action

☒ View for all users

☐ Generate notification

Apply

Cancel

Figure 36 – Rebuild Historical data: Dates setting

1.2.1.5 Example

Before calibrating the particles, we need to study their behavior to detect a specific period where we know how they should behave.

Let us suppose we want to calibrate the particle concentration of an air quality device tagged as 'K-A14 Device 15'. This device has the following PM historical data (Data > Standard):

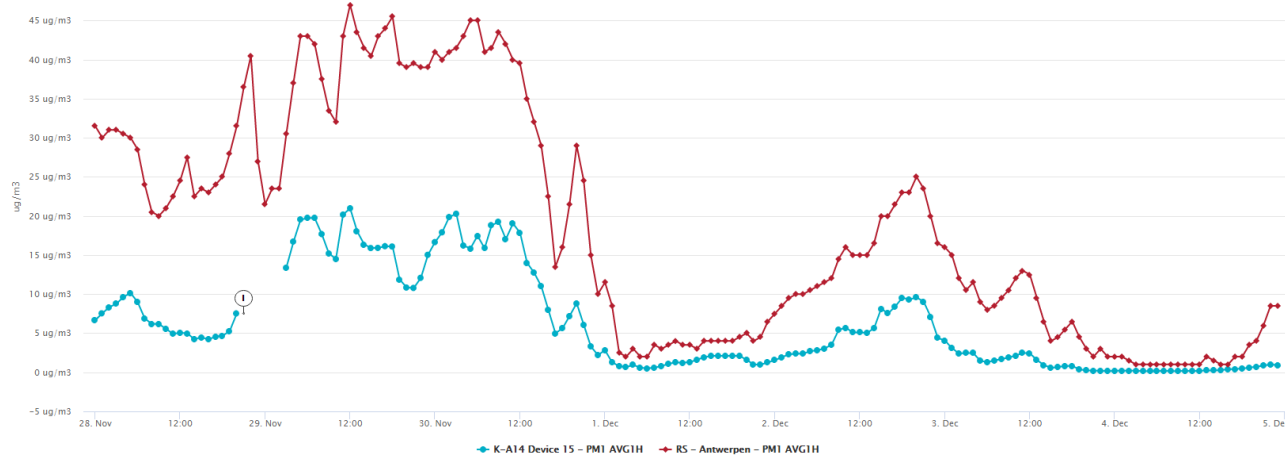


Figure 37 – Particle calibration: PM₁ example

We are going to use another device as a reference ('RS - Antwerpen'). This device should be placed in the same location as the one we want to calibrate.

Once the preliminary concentration analysis is performed, you can start with the baseline calibration from "Operation" tab > "Gas & Particle Calibration" > "Particle calibration".

The first step is to select the 'K-A14 Device 15" device and the NO₂ sensor in the multiple devices and single gas selectors:

Particle Calibration

1
Calibration Type & Device

2
Check data

3
PM factors

4
Show result

Step 1 - Calibration Type & Device

Particle Calibration Type

Lineal Regression (forced origin) against ▼

Device: K-A14 Device 15 ▼

Previous

Next

Figure 38 – Particle calibration example: select Particle Calibration Type and device

Then, we choose the dates from 27th November 2020 until 4th of December 2020, and the RS – Antwerpen as reference station for the PM₁₀, PM_{2.5} and PM₁. Thus, we select the RS – Antwerpen in the “Reference Device with PM10” selecting box, and then, “PM10” as “Parameter PM10”. The same steps are done for PM_{2.5} and PM₁.

Particle Calibration

✓ 1
Calibration Type & Device

2
Check data

3
PM factors

4
Show result

Step 2 - Check data

Period to be charted and if required to rebuild

Time zone
(UTC+1:00) Brussels, Copenhagen, Madrid, Paris ▼

Start date
2020-11-28 

End date
2020-12-04 

Start time
00:00 

End time
23:59 

Reference device(s) 

Reference Device with PM10 RS - Antwerpen ▼	Parameter PM10: PM10 ▼
Reference Device with PM2.5 RS - Antwerpen ▼	Parameter PM2.5: PM2.5 ▼
Reference Device with PM1 RS - Antwerpen ▼	Parameter PM1: PM1 ▼

Previous

Next

Figure 39 – Particle calibration example: select calibration period and check data period

The new PM factors are calculated regarding the period selected in the step before. The “New factor” calculated in the slope forced to the origin, thus the “New offset” value will remain as 0.

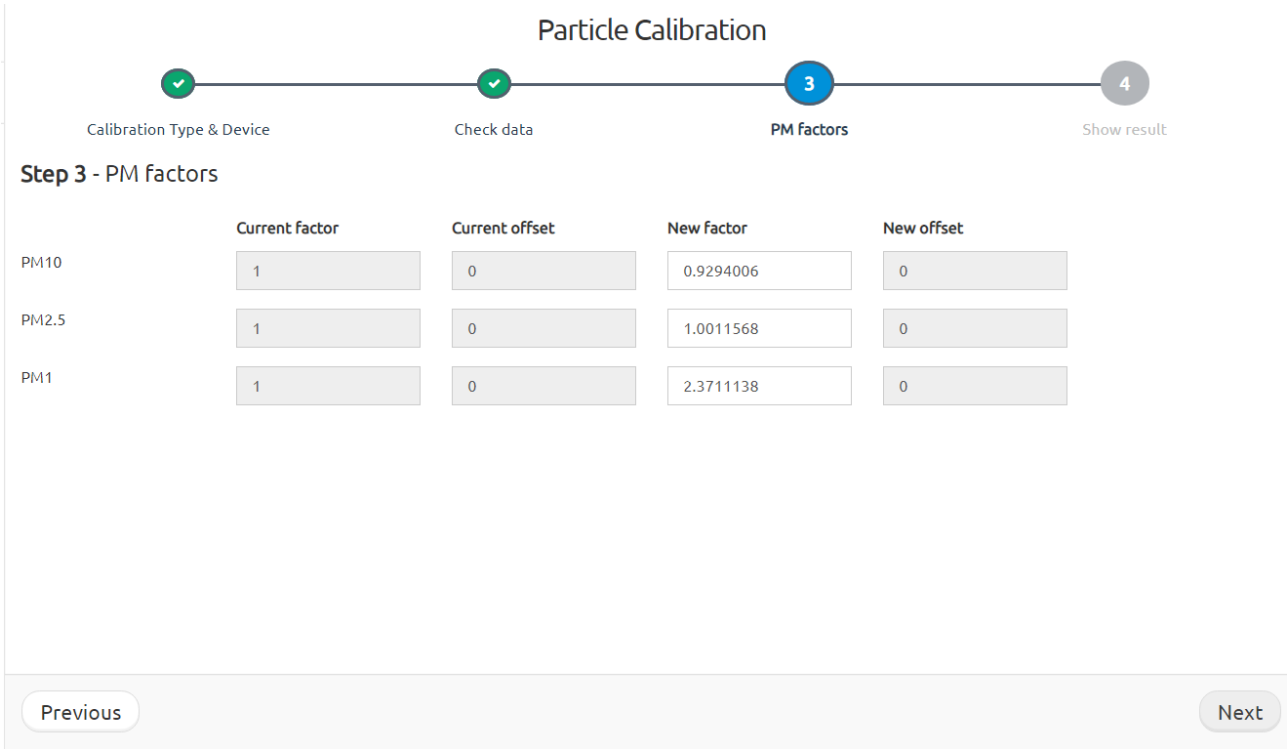


Figure 40 – Particle calibration example: PM factors

In step 4 the results obtained from this calibration will be shown:

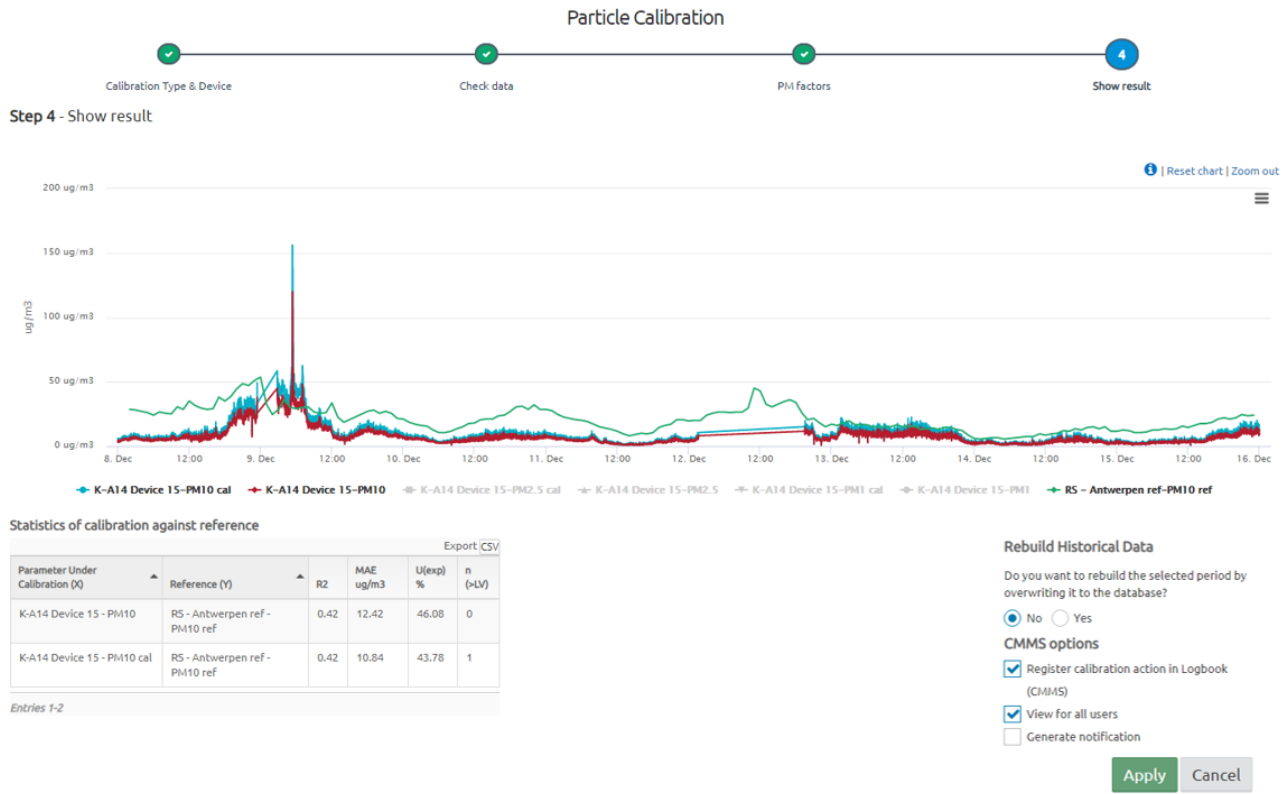


Figure 41 – Particle calibration example: graph with the new calibration results from PM10

The calibration seems to be correct so now we can “Rebuild the Historical Data” by clicking on the “Yes” option and selecting the dates desired to rebuild. Select as required the different “CMMS options”. Finally, apply the calibration by clicking on the “Apply” button.

After confirming the calibration, the new settings will be queued and applied to start from the next communication period.

1.2.2 Manual

The “Manual” option allows to insert calculated PM factors, externally from the Cloud, and calculate the basic statistics with the new factors as well as apply them.

Follow the steps below to perform a correct “Manual” particle calibration through the **KunakCloud** web.

1.2.2.1 Calibration Type and Device

The very first step in a calibration process is the type of Particle Calibration, in this case, “Linear Regression (forced origin) against reference” and then select the device to calibrate. To go to the next step, click on the “Next” button.

As an example, we use the device K-A14 Device 15.

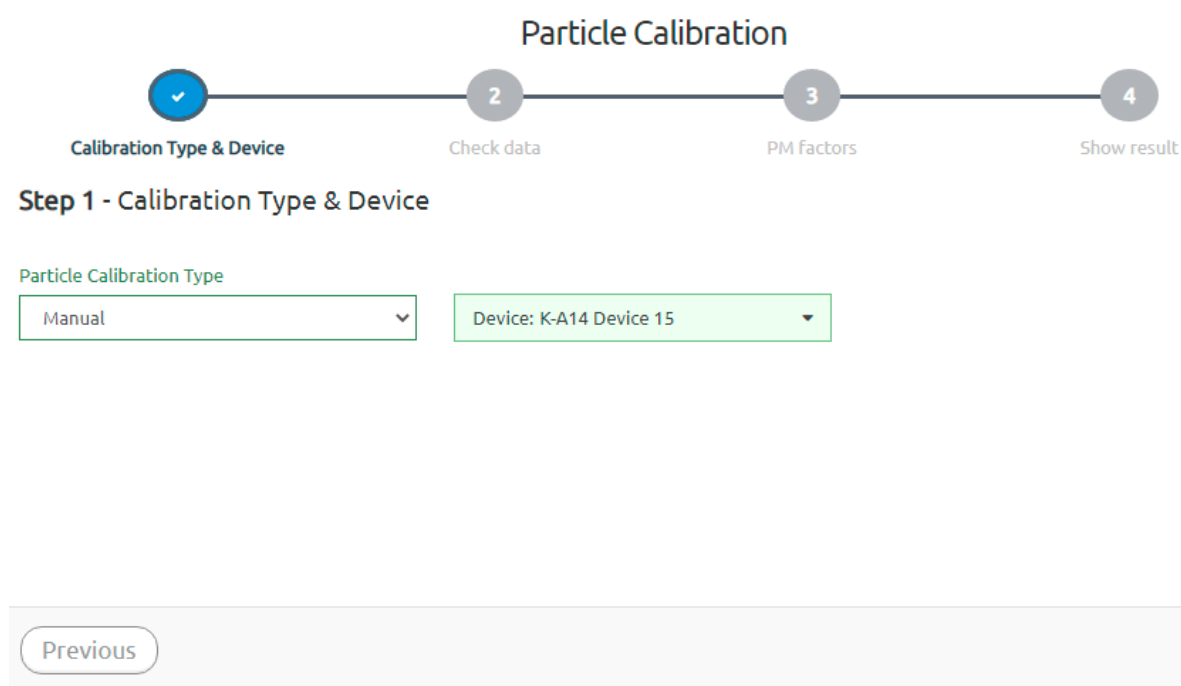


Figure 42 – Manual particle calibration tool.

1.2.2.2 Check data

In this step, you must select the period to be charted and if required to rebuild. The results of your calibration in the existent data before applying the calibration will be shown. Optionally, it exists 3 reference devices to select, one for each particle size (PM10, PM2.5 and PM1).

As an example, we choose the dates from 27th November 2020 until 4th of December 2020, and the RS – Antwerpen as reference station for the PM10, PM2.5 and PM1. Thus, we select the RS – Antwerpen in the “Reference Device with PM10” selecting box, and then, “PM10” as “Parameter PM10”. The same steps are done for PM2.5 and PM1.

To go to the next step, click on the “Next” button.

Particle Calibration

✓
 Calibration Type & Device

2
Check data

3
 PM Factors

4
 Show result

Step 2 - Check data

Period to be charted and if required to rebuild

Time zone
 (UTC+1:00) Brussels, Copenhagen, Madrid, Paris

Start date: 2020-11-28 Start time: 00:00

End date: 2020-12-04 End time: 23:59

Reference device(s) i

Reference Device with PM10 RS - Antwerpen	Parameter PM10: PM10
Reference Device with PM2.5 RS - Antwerpen	Parameter PM2.5: PM2.5
Reference Device with PM1 RS - Antwerpen	Parameter PM1: PM1

Previous
Next

Figure 43 - Particle calibration: check data

1.2.2.3 PM Factors

The manual PM factors are added in the “New factor” and “New offset” windows.

In the first two columns, the “Current factor” and “Current offset” values appear in case another calibration was performed before. Thus, if not prior calibration was carried out, the “Current factor” and “Current offset” will be 1 and 0 respectively, as shown in Figure 44 – Particle calibration: Manual PM factors and offsets (not previous calibration). If a calibration has been performed previously, the “Current factor” and “Current offset” will show the values applied in the last calibration, as shown in Figure 45 – Particle calibration: Manual PM factors and offsets (previous calibration performed). The “Current factors” will be deleted, and substitute by the “New factor”(s) calculated.

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1	0	0.95	1.2
PM2.5	1	0	0.82	0.96
PM1	1	0	0.70	0.5

Previous
Next

Figure 44 – Particle calibration: Manual PM factors and offsets (not previous calibration)

Particle Calibration

✓

✓

3

4

Calibration Type & Device
Check data
PM factors
Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1.429646	0	1.3192322	1.7
PM2.5	1.296212	0	1.25	0.95
PM1	1.296212	0	1.13	0.5

Previous
Next

Figure 45 – Particle calibration: Manual PM factors and offsets (previous calibration performed)

A constraint appears if PM10 factor is lower than PM2.5 and PM1, and if PM2.5 factor is lower than PM1. For more details see the *Lineal Regression (forced origin) against reference* Section.

To select the “New factors” click on “Next” button.

1.2.2.4 Show result

Before applying the calibration, the chart specified in the previous step will be plotted showing how the calibration will be. The curves that will appear in the chart shown are:

- **Blue** line (K-A14 Device 15 – PM10 cal) is the calibrated concentration. The tag follows the format '{calibrated device name}-{particle name} cal'.
- **Red** line (K-A14 Device 15 – PM10) is the original concentration. The tag follows the format '{calibrated device name}-{gas name}'.
- **Green** line (RS Antwerpen ref-PM10 ref) is the reference sensor. Up to three reference sensors, one for each particle size, can be selected and they will be tagged as '{reference device name}-{ sensor name} ref'.

The statistics of calibration against reference are calculated with the original data and with the calibrated data. It shows the R2, the Minimum Absolute Error (MAE), the expanded uncertainty (U(exp)%) and the values above the Limit Value set by the Data Quality Objectives of the European Directive 2008/50/EC.

If the values of the chart are correct, it exists the possibility to “Rebuild Historical data”, as appeared in by overwriting the data with the new calibrated values over the selected period by clicking on the “Yes” option when asking to rebuild the data. Only in the Particle Calibration tool, this option gives you the option to select the start and end date that you would like to rebuild data, as shown in Figure 36 – Rebuild Historical data: Dates setting. Besides, you have the possibility to register the calibration in the CMMS. For more details see the *Baseline calibration* Section.

Finally, by clicking on the “Apply” button a confirmation dialog message will be shown. Clicking on “YES”, a completed message will appear at the top-right of the web. The new settings will be queued and applied to start from the next communication period.

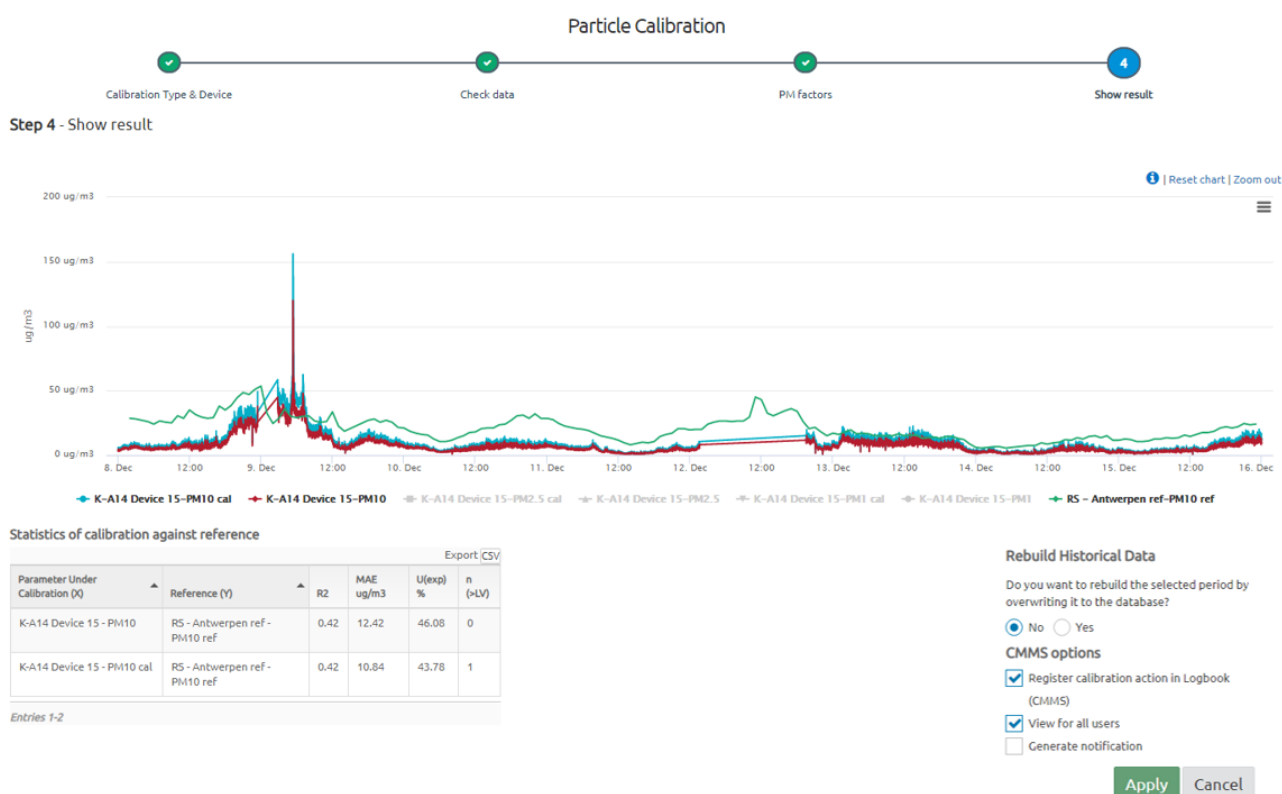


Figure 46 – Particle calibration: show results

Rebuild Historical Data

Do you want to rebuild the selected period by overwriting it to the database?

☐ No ☒ Yes

Start date

2021-01-31



Start time

00:00



End date

2021-02-04



End time

23:59



CMMS options

☒ Register calibration action in Logbook (CMMS)

☐ Add the statistics in the Logbook action

☒ View for all users

☐ Generate notification

Apply

Cancel

Figure 47 – Rebuild Historical data: Dates setting

1.2.2.5 Example

The same example used for the Lineal Regression (forced origin) against reference is used in this example.

Before calibrating the particles, we need to study their behavior to detect a specific period where we know how they should behave.

Let us suppose we want to calibrate the particle concentration of an air quality device tagged as 'K-A14 Device 15'. This device has the following PM historical data (Data > Standard):

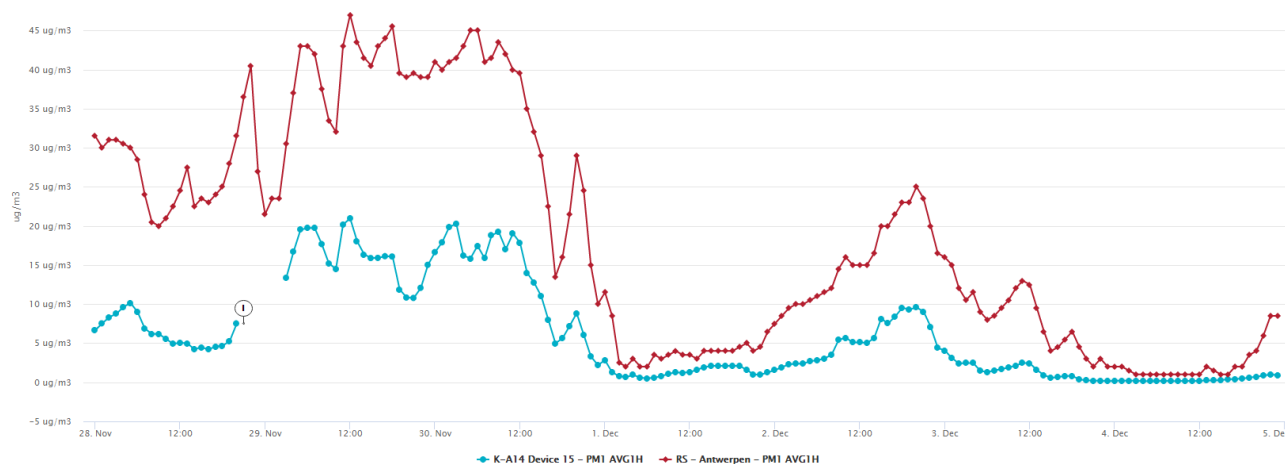


Figure 48 – Particle calibration: PM₁ example

We are going to use another device as a reference ('RS - Antwerpen'). This device should be placed in the same location as the one we want to calibrate.

Once the preliminary concentration analysis is performed, you can start with the baseline calibration from "Operation" tab > "Gas & Particle Calibration" > "Particle calibration".

The first step is to select the 'K-A14 Device 15" device and the NO2 sensor in the multiple devices and single gas selectors:

Particle Calibration

1
 Calibration Type & Device

2
 Check data

3
 PM Factors

4
 Show result

Step 1 - Calibration Type & Device

Particle Calibration Type

Lineal Regression (forced origin) against ▼

Device: K-A14 Device 15 ▼

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Figure 49 – Particle calibration example: select Particle Calibration Type and device

Then, we choose the dates from 27th November 2020 until 4th of December 2020, and the RS – Antwerpen as reference station for the PM₁₀, PM_{2.5} and PM₁. Thus, we select the RS – Antwerpen in the “Reference Device with PM10” selecting box, and then, “PM10” as “Parameter PM10”. The same steps are done for PM2.5 and PM1.

In this case, the selected period is only used for checking the data, not for obtaining the new factors because they will be calculated by the user and include manually.

Particle Calibration

✓
 Calibration Type & Device

2
 Check data

3
 PM Factors

4
 Show result

Step 2 - Check data

Period to be charted and if required to rebuild

Time zone
 (UTC+1:00) Brussels, Copenhagen, Madrid, Paris ▼

Start date
 2020-11-28

Start time
 00:00

End date
 2020-12-04

End time
 23:59

Reference device(s) ?

Reference Device with PM10 RS - Antwerpen ▼	Parameter PM10: PM10 ▼
Reference Device with PM2.5 RS - Antwerpen ▼	Parameter PM2.5: PM2.5 ▼
Reference Device with PM1 RS - Antwerpen ▼	Parameter PM1: PM1 ▼

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Figure 50 – Particle calibration example: select calibration period and check data period

This step allows to include the “New factor” and “New offset” manually

Particle Calibration

✓

✓

3

4

Calibration Type & Device

Check data

PM factors

Show result

Step 3 - PM factors

	Current factor	Current offset	New factor	New offset
PM10	1	0	0.95	1.2
PM2.5	1	0	0.82	0.96
PM1	1	0	0.70	0.5

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Figure 51 – Particle calibration example: PM factors

In step 4 the results obtained from this calibration will be shown:

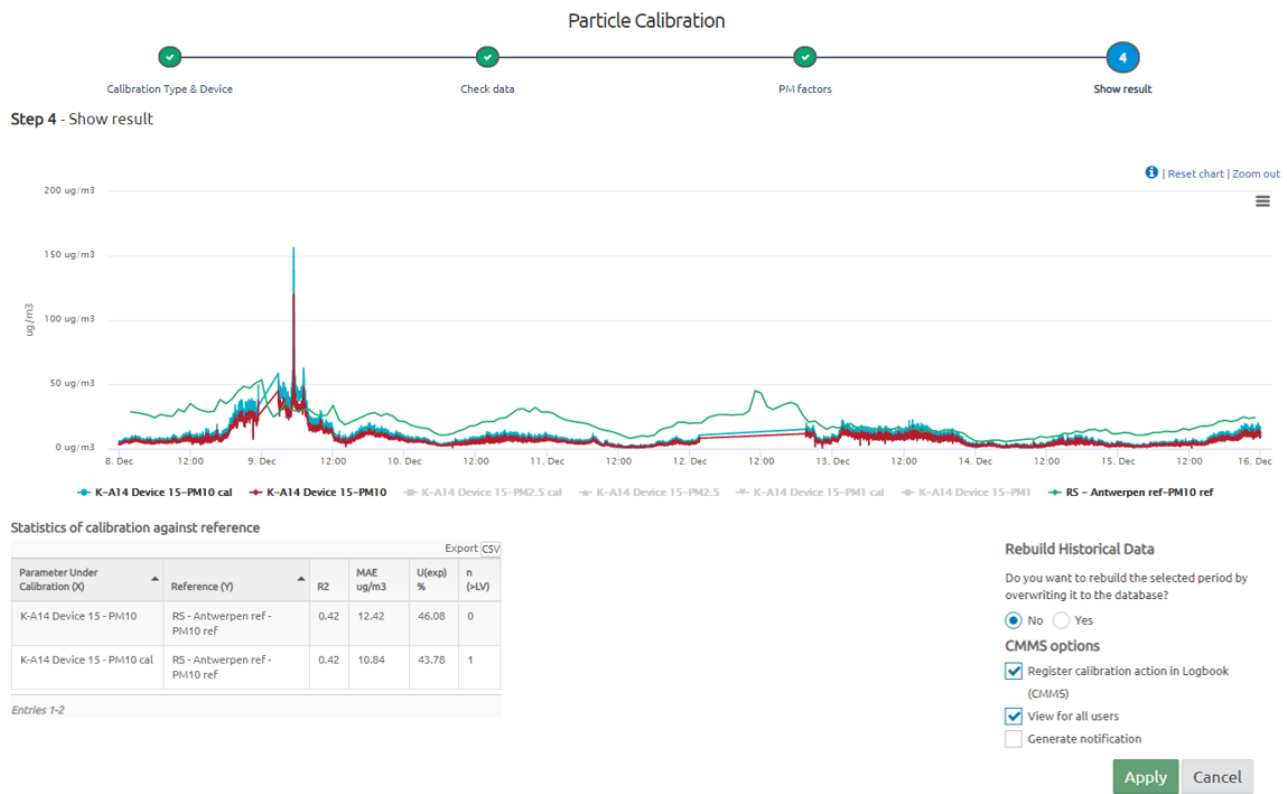


Figure 52 – Particle calibration example: graph with the new calibration results from PM10

The calibration seems to be correct so now we can “Rebuild the Historical Data” by clicking on the “Yes” option and selecting the dates desired to rebuild. Select as required the different “CMMS options”. Finally, apply the calibration by clicking on the “Apply” button.

After confirming the calibration, the new settings will be queued and applied to start from the next communication period.